

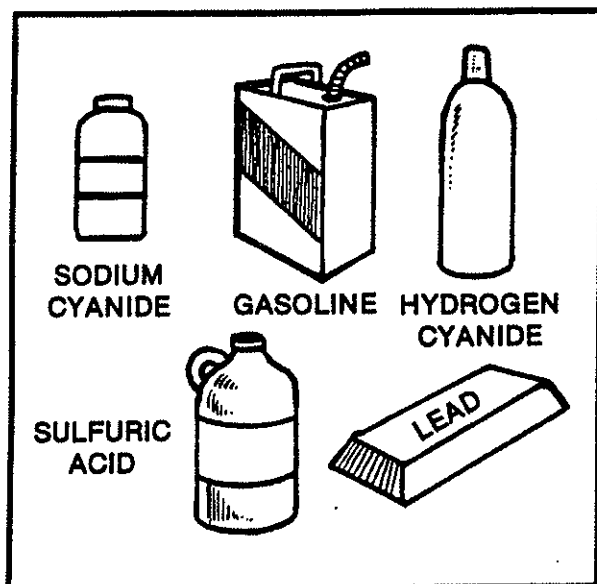
HAZARDOUS CHEMICAL SAFETY



J.T.Baker

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Phillipsburg, NJ 08865

3. Toxic Chemicals



What is a Toxic Chemical?

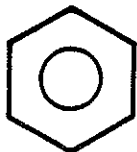
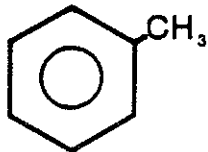
Any chemical which, according to Dorland's Medical Dictionary, "when ingested, inhaled, or absorbed, or when applied to, injected into or developed within the body, in relatively small amounts, by its chemical action may cause damage to structure or disturbance to function."

What Type of Damage Do They Cause?

Toxic substances may cause injury by damaging biological structure and/or disturbing biological function.

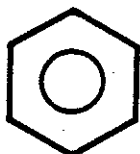
SOME COMMON CHEMICALS CLASSIFIED BY TYPE OF DAMAGE		
Damage to Structure	Disturbance of Function	Both Effects
Sulfuric acid, 50% sodium hydroxide, and other corrosive materials are discussed in Chapter 9.	Carbon monoxide decreases the ability of hemoglobin to transport oxygen to cells. Parathion is a cholinesterase inhibitor; thus it blocks the transfer of nerve impulses.	Chlorine when inhaled damages the lungs; this damage can inhibit the oxygen exchange in the lungs.



BENZENE**TOLUENE****HAZARD RATINGS***

Acute Systemic:	2	2
Chronic Systemic:	3	2
TWA	10 ppm	100 ppm

*See Chapter 7

**BENZENE**

ACUTE: Irritation, aromatic
hydrocarbon poisoning

CHRONIC: Anemia, leukemia,
death

Damage can be local and/or systemic:

Local toxicity may be defined as the effect of a substance on the body area that has been exposed to the substance. Exposure may be through direct contact, inhalation, ingestion, or penetration.

Systemic toxicity is the effect of a substance on body tissues after absorption into the bloodstream. Absorption may take place through the skin, stomach, or lungs.

Damage can be acute and/or chronic:

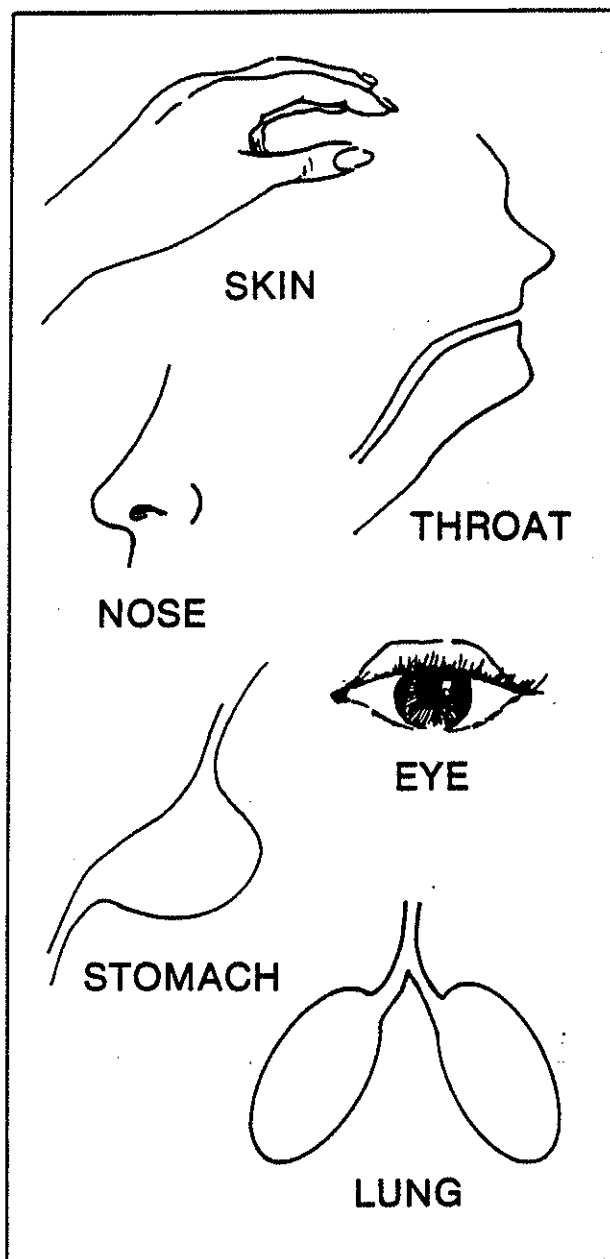
Two substances may have about the same toxicity on a single exposure; however, their effect on prolonged or repetitive exposure may be quite different. In other words, the toxic nature of a substance must be related to the duration of exposure.

ACUTE TOXICITY—Is usually defined as that effect manifested on "short exposure," on single contact, ingestion, or inhalation.

CHRONIC TOXICITY—Is that effect observed when a toxicant acts on body tissues over a "long" period of time, days to years. A chronically toxified subject may not observe effects until an advanced stage is reached and permanent damage is done.



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What Are Some of the Parameters Which Affect Toxicity?

- A. **Mode of Entry**—Commonly, toxic materials contact the skin, eye, respiratory tract, and/or digestive system.
- B. **Physical Condition**—Are you ill? Has your body been damaged by a previous assault?
- C. **Dose**—How much of the chemical assaulted your system?
- D. **Duration**—How long were you exposed to the chemical?
- E. **Sensitivity**—How sensitive are you to the chemical?
- F. **Combined Effects**—What other chemicals were you exposed to?
- G. **Stress**—Were you under any physical or mental stress during or after the exposure?
- H. **Others**—Sex, race, temperature, altitude, and a host of additional parameters sometimes affect toxicity.



What Amount of a Particular Chemical Is Toxic?

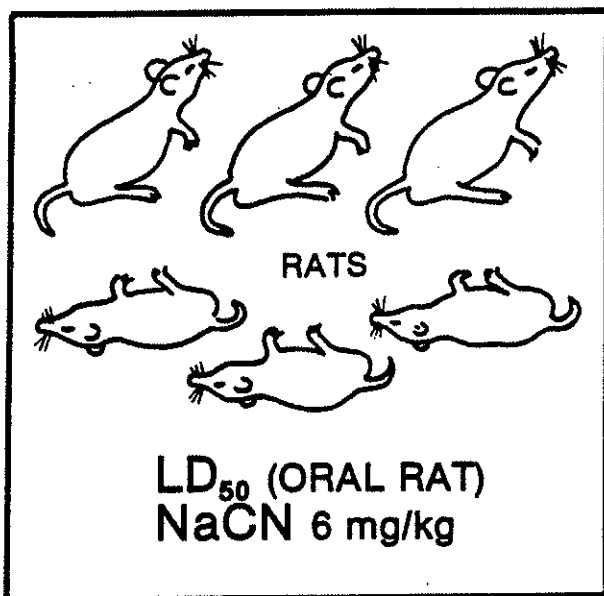
All chemicals can be toxic! A gram of salt will kill a rat; five grams of caffeine will kill a human; a single tablet of penicillin can kill people with allergies.

TOXIC AND NORMAL LEVELS OF CHEMICALS IN THE BLOOD		
Chemical	Normal Level ¹ Milligrams %	Toxic Level ² Milligrams %
Acetone	0	30
Acetylsalicylic Acid (aspirin)	5	40
Arsenic	< 0.002	1
Benzene	0	any trace
Bromide	5	180
Carbon Monoxide	1	30
Cyanide	0.01	0.2
Ethanol	0	250
Ether	100	150
Fluoride	< 0.05	0.1
Lead	< 0.01	0.1
Oxalic Acid	0.2	1.0

¹Normal levels include therapeutic levels in the case of these chemicals.

²These are the levels which may be fatal. Discernible toxicity usually occurs at lower levels.



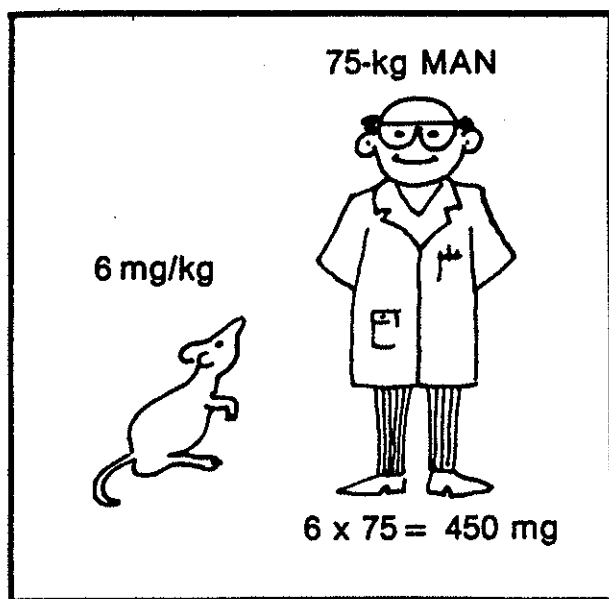


How Are Toxic Doses Determined?

LD₅₀ (Lethal Dose, 50% Kill)—Is the dose which, when administered to test animals, kills half of them. The LD₅₀ is expressed in mg of toxicant per kilogram of body mass, and the test route is given (oral, intravenous, etc., but not inhalation).

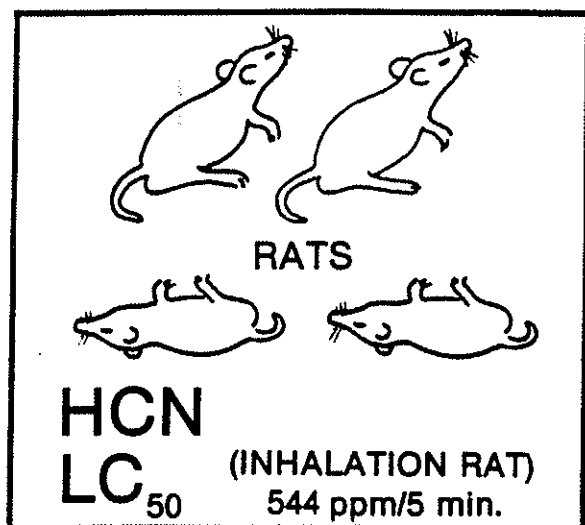
LD_{L0} (Lethal Dose, Low)—Is the lowest dose over any given period of time in either single or divided portions which has caused death in humans or animals. The exposure route is specified and may be anything other than inhalation.

LD₁, LD₁₀, LD₃₀, and LD₉₉ are sometimes published in the literature. They present the lethal dose to kill 1%, 10%, 30% and 99% of the test animals, respectively.



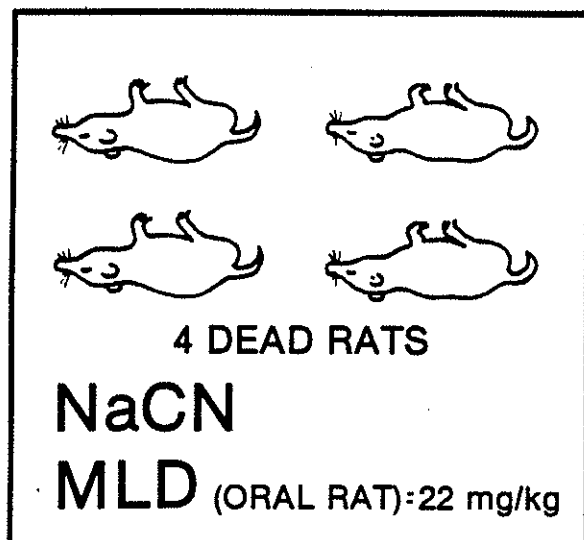
How does the LD₅₀ for rats relate to man? If a toxicant, say, sodium cyanide, is as toxic to man as it is to rats, the LD₅₀ is identical. By extrapolation, assuming a LD₅₀ of 6 mg/kg for both rats and humans, a 450-mg dose of sodium cyanide would kill a human if administered orally. Evidence shows that 200 mg of sodium cyanide can kill a 75-kg man under these circumstances. Species differences make such direct extrapolation very tenuous.





LC₅₀ (Lethal Concentration, 50% Kill)—Is that concentration of a toxicant in air or other suitable vehicle, which, when administered to test animals for a definite time period, kills 50% of them. The LC₅₀ is given in parts per million (ppm) for that time period.

LC_{L0} (Lethal Concentration, Low)—Is the lowest concentration which has been reported to have caused death in humans or animals.



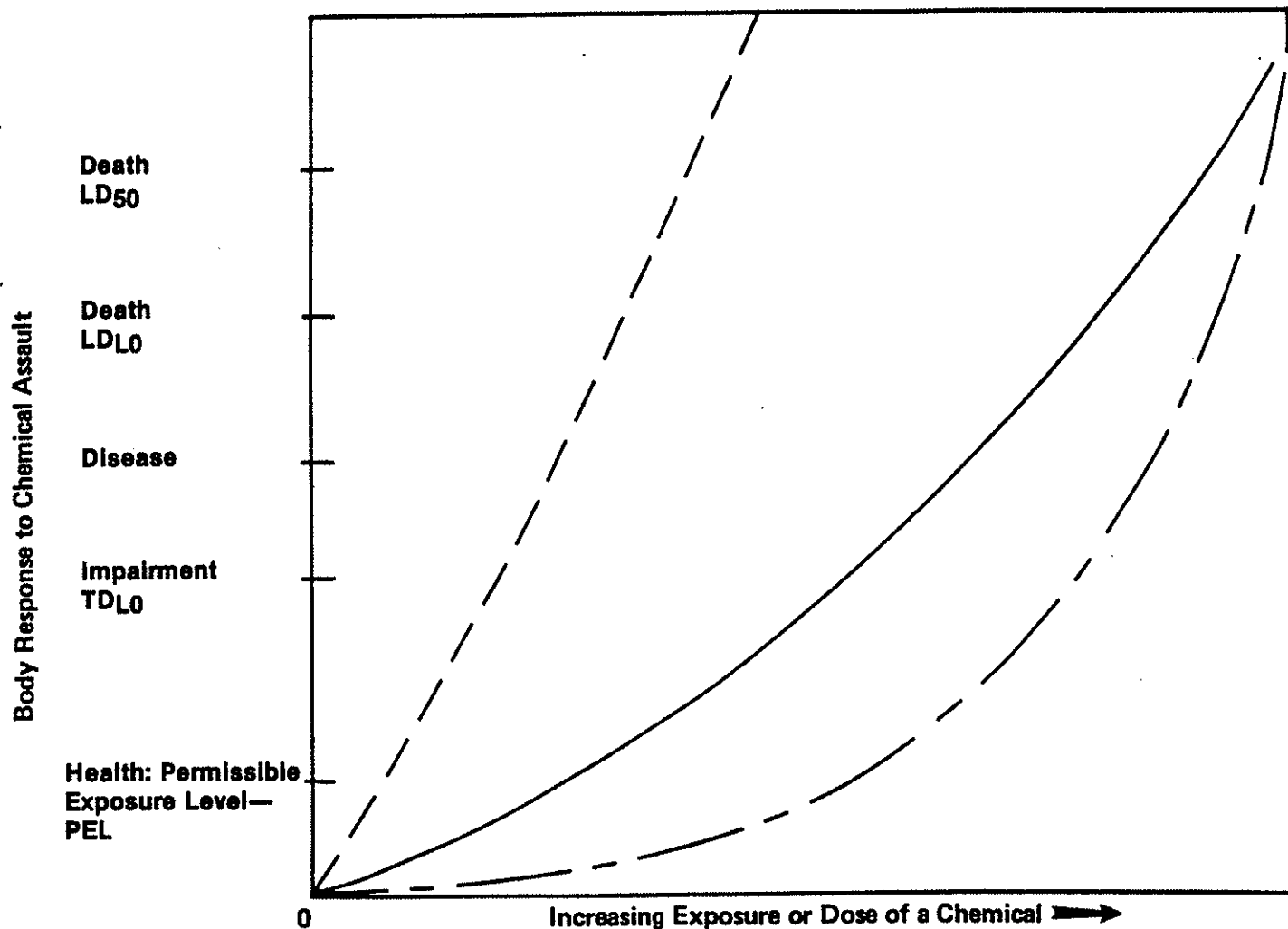
MLD (Minimum Lethal Dose)—Is an obsolete term, usually found only in older literature, and represents the dosage at which all or nearly all of the test animals are killed.

TD_{L0} (Toxic Dose, Low)—Is the lowest dose of a substance introduced by any route, other than inhalation, over any given period of time and reported to produce carcinogenic, neoplastigenic, teratogenic, or mutagenic effects in humans or animals, or to produce any toxic effects in humans.



How Can Safe Exposure Levels for Humans Be Determined?

- A. By extrapolating acute toxicological data such as LD₅₀ values.
- B. By studying human exposure data.
- C. By drawing analogies.
- D. By assuming that there exists some permissible exposure level below which no impairment will occur no matter how frequent the exposure.



THRESHOLD LIMIT VALUE**Benzene****10 ppm****Sodium
Cyanide****5 mg/m³****Why the TLV should not be applied literally**

1. Based on limited experimentation in many cases
2. Concentration of toxicants changes with time and may exceed TLV
3. The TLV may not apply to hyper-sensitive individuals
4. Industrial exposure may involve mixtures about which there may be little information as to toxicity

TLV* (Threshold Limit Value)—is an estimate of the average safe toxicant concentration that can be tolerated on a repetitive basis, usually an eight-hour period on a day-to-day basis. The units of TLV are ppm or mg/m³. The TLV is usually applied to airborne contaminants (gases, mists, dust, etc.). Formerly, the TLV was referred to as the MAC, or Maximum Allowable Concentration (also expressed in ppm or mg/m³). The MAC is an estimate of toxicant concentration below which ill effects are unlikely to be experienced. See Chapter 10 of this manual.

The TLV's should not be applied literally. They should be used as general guidelines when working with toxicants and as engineering parameters.

TLV's* are established by the American Conference of Governmental Industrial Hygienists; the term is a registered trademark. In past years the following procedures were used to establish TLV's:¹

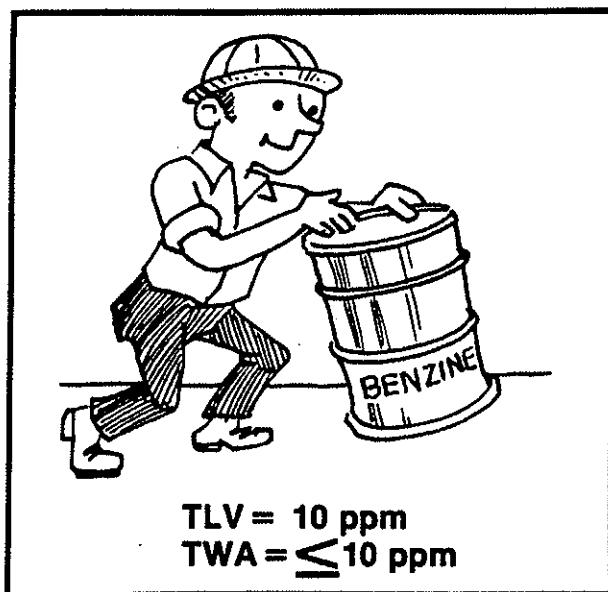
- A. Human experience in industry, 38%.
- B. Human volunteer experiments, 11%.
- C. Chronic animal inhalation studies, 20%.
- D. Acute animal inhalation studies, 2%.
- E. Animal ingestion studies, 5%.
- F. Analogy, 24%.

¹Occupational Safety and Health Series No. 20, International Labor Office, Geneva, Switzerland, 1970.



$$TWA = \sum_{i=1}^{i=n} \frac{T_i C_i}{T_{\text{total}}}$$

$$= \frac{\left(\begin{matrix} \text{exposure} \\ \text{time A} \end{matrix} \right) \left(\begin{matrix} \text{conc} \\ \text{A} \end{matrix} \right) + \left(\begin{matrix} \text{exposure} \\ \text{time B} \end{matrix} \right) \left(\begin{matrix} \text{conc} \\ \text{B} \end{matrix} \right) + \dots}{\text{total time worked per shift}}$$



TLV® -TWA (Time Weighted Average)—Is the sum of the products of the toxicant concentration and duration of exposure, taken over the total exposure time. The concentration of a toxicant is multiplied by the total time of that exposure. The concentration during a second time period is multiplied by the time of that exposure. The products are added and then divided by the time worked per shift.

The TWA is used to assess the average exposure during a working shift, thus allowing this average to be related to the TLV for the relative assessment of safe working conditions. Depending on sampling, the TWA may or may not reveal "surges" above the TLV.

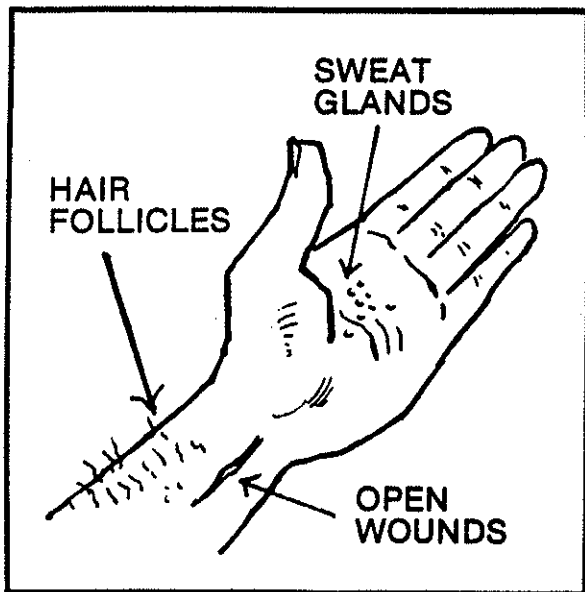
TLV® -C (Ceiling Value)—Is the concentration which should not be exceeded even instantaneously.

TLV® -STEL (Short Term Exposure Limit)—Is the maximum exposure during 15 minutes.

PEL (Permissible Exposure Limit)—Is the maximum exposure recommended by NIOSH. These are frequently derived from the TWA values given by ACGIH and are ultimately enforced by OSHA.

IDLH (Immediately Dangerous to Life and Health)—Is a term used by NIOSH to define maximum allowable short-term exposures.



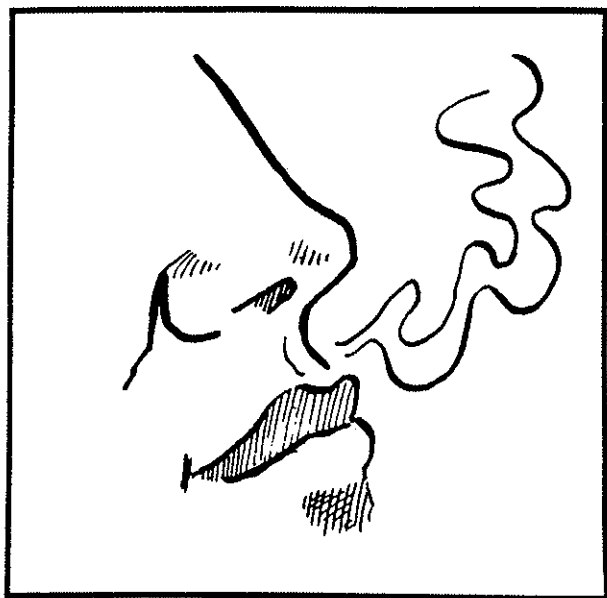


What Are the Common Modes of Entry for Toxic Chemicals?

SKIN—The skin is the most frequently exposed body tissue to toxicants. The most common result is irritation. Many toxicants, however, are rapidly absorbed through the skin, and may cause systemic effects.

The points of entry in the skin are hair follicles, sweat glands, and open wounds. Rapid distribution of toxicants is facilitated by the lymphatic system.

NOTE: Toxicant contamination of clothing and shoes increases the hazard because of the localization of toxic concentrations. Pain sensations do not necessarily accompany an assault on the skin.



NOSE (Inhalation)—The internal surface area of the lungs is estimated to be greater than 100 m², thus allowing rapid absorption of toxicants into the bloodstream. Absorption rate and activity (increasing the respiration rate) are related—the more active the person, the faster the absorption. This is the mode of entry of most concern to the industrial hygienist.

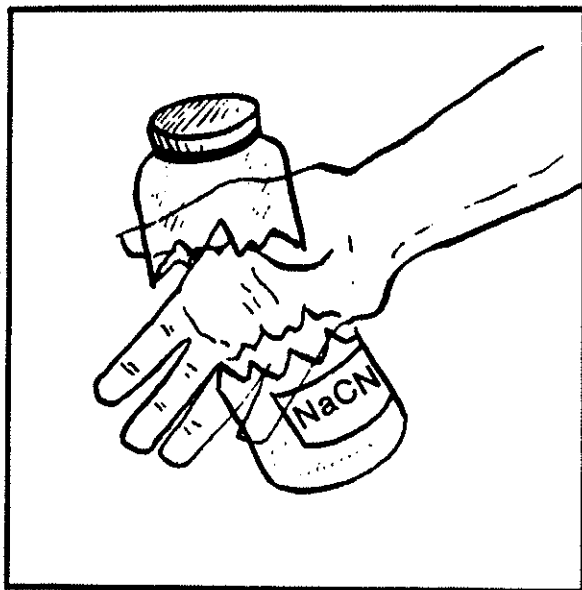
Many toxicants reveal their presence by giving sensory warnings, such as pain or smell. These sensory effects can be useful in this respect, because immediate remedial action can be taken. Often, however, these senses may become aware of danger after the fact, or may not respond at all. Olfactory fatigue may prevent observation of toxic quantities of O₃ or H₂S, for instance. Or the pain from a hydrofluoric acid burn may not be noticed until several hours have passed.

Sensory warning is a useful backup, but should not be relied on as a first defense.





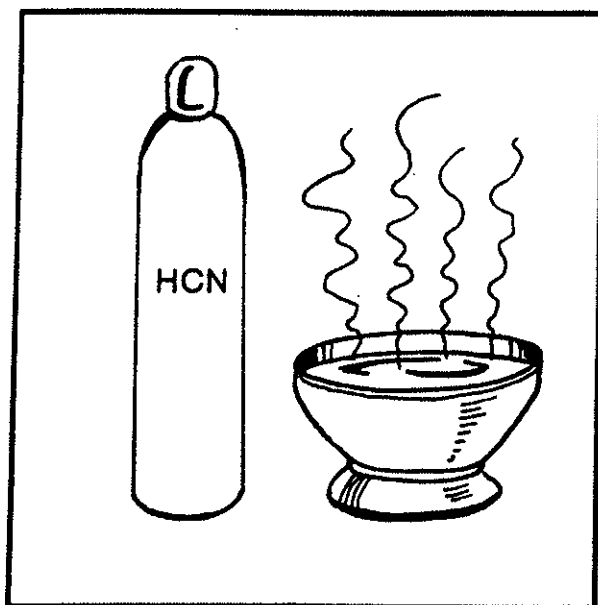
EYES—Very few substances are safe or innocuous when in contact with the eyes. The sensitivity of the eye to toxicants (chemicals in general) is such that irritation, pain, impairment of vision, or even loss of vision can result.



INGESTION—Since only minute amounts of toxicants can be ingested by swallowing contaminated air, the primary routes are purposeful or accidental ingestion of copious amounts of the toxicants.

INJECTION—The equivalent of purposeful injection of toxicants can occur through mechanical injury with contaminated glass or metal—a cut.

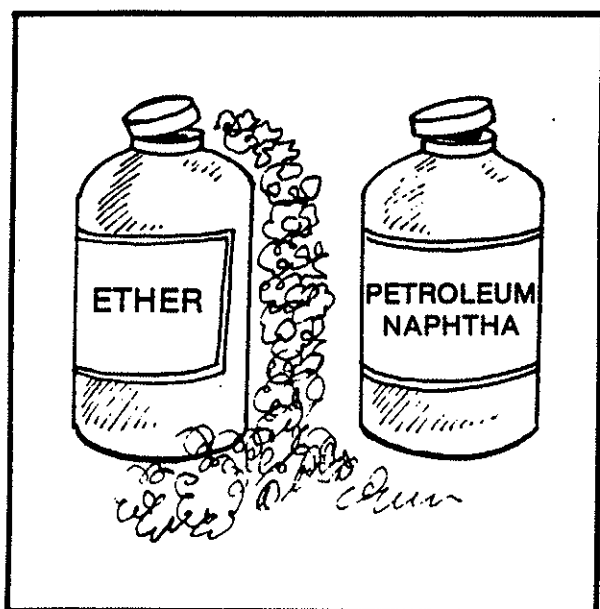




How Are the Physical Properties and Toxicity of Chemicals Related?

Relative Hazards—A sealed container of hydrogen cyanide represents less hazard than an open vessel of benzene, simply because the chances of exposure are greater for the benzene. All potential toxicants should be prevented from entering your biosystem.

When assessing relative toxicity, keep in mind that gases are absorbed more rapidly than liquids, which in turn, are absorbed more rapidly than solids.



Physical Properties—A substance with a boiling point of, say, 40°C, is obviously potentially more contaminating than one boiling at 200°C.

NOTE: The control of toxic substances must be based on a sound knowledge of the physical properties of those substances, and on the knowledge of how to handle and store them. All chemicals can be handled safely.

KNOWN CARCINOGENS TO MAN

4-Aminobiphenyl
 Analgesic mixtures containing phenacetin
 Arsenic and certain arsenic compounds
 Asbestos
 Azathioprine
 Benzene
 Benzidine
 N, N-bis (2-chloroethyl)-2-naphthylamine
 (chlornaphazine)
 Bis (chloromethyl) ether and technical grade
 chloromethyl methyl ether
 1,4-Butanediol dimethylsulfonate (myleran)
 Certain combined chemotherapy for
 lymphomas
 Chlorambucil
 Chromium and certain chromium compounds
 Coke oven emissions
 Conjugated estrogens
 Cyclophosphamide
 Diethylstilbestrol
 Hematite underground mining
 Isopropyl alcohol manufacturing (strong
 acid process)
 Manufacture of auramine
 Melphalan
 Methoxsalen with ultra-violet A therapy (PUVA)
 Mustard gas
 2-Naphthylamine
 Nickel refining
 Rubber industry (certain occupations)
 Soots, tars, and mineral oils
 Thorium dioxide
 Vinyl chloride

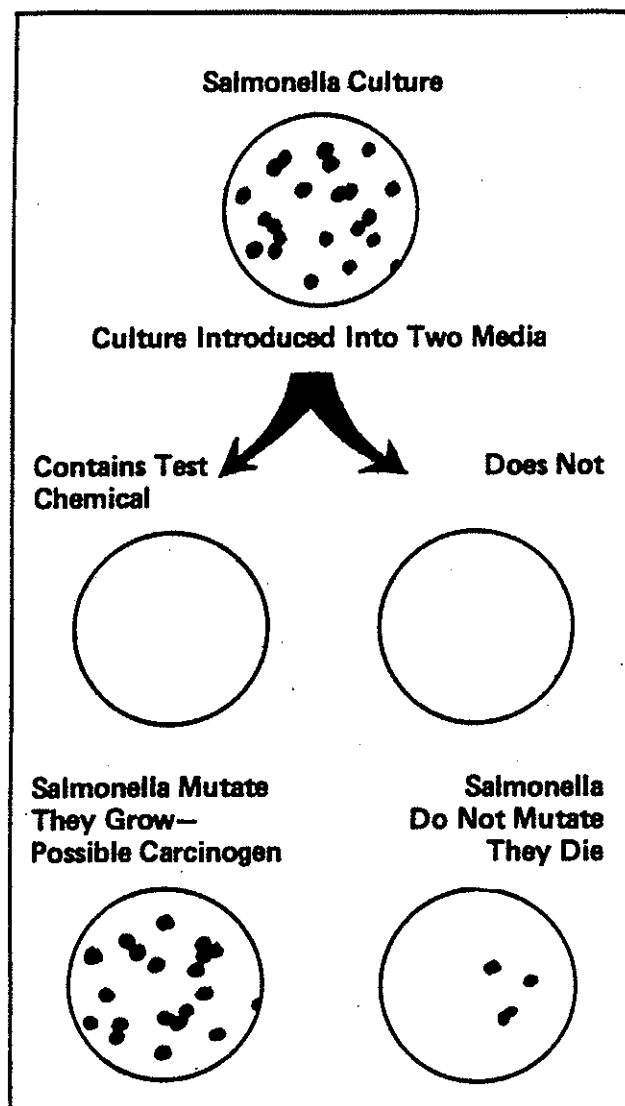
What About Long-Term-Exposure and Delayed-Response Studies?

Carcinogens In Man—A carcinogen is a substance that will induce a malignant tumor in man following a reasonable exposure. The tumor can form quickly or after a delay of more than a decade. Typical known carcinogens in man are listed in the table to the left.

Mutagenesis In Man—A mutagen is a substance that affects the genetic system of the exposed person in such a way that it may cause hereditary changes in the chromosomes. Such genetic damage cannot be detected until children are produced and may even be delayed for several generations. No clearly definable chemical mutagens in man have been identified.

From NTP



AMES' SALMONELLA TEST

Bruce Ames has developed this test which correlates mutagenesis of a strain of bacteria (Salmonella) with carcinogenesis in animals, including man.

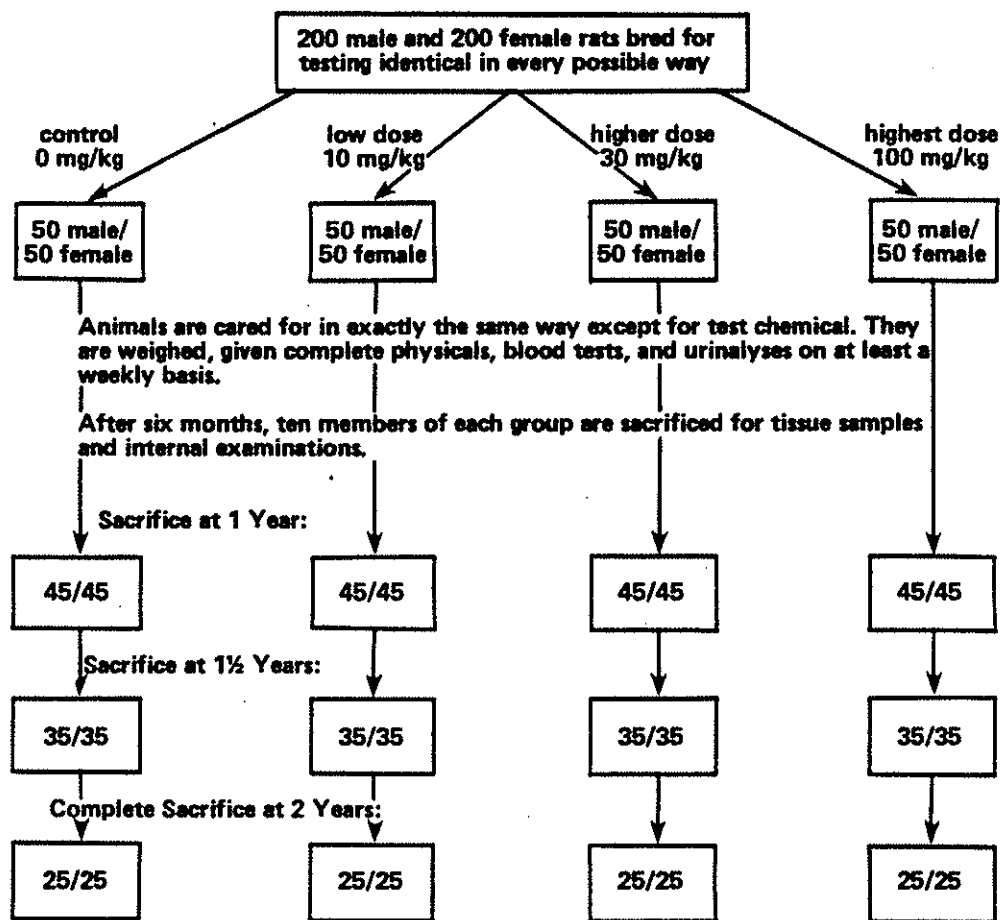
Teratogenesis in Man—A teratogen is a substance which interferes with the normal development of the fetus without inducing damage to the mother or killing the fetus. The deformation is not hereditary; it is physical in nature. Thalidomide is a confirmed teratogen in man; steroids may possibly be.

Suspected Carcinogens, Mutagens, and Teratogens in Man—No single test is available, but animal tests continue to be most acceptable. To screen chemicals as potential carcinogens using animals can be prohibitively expensive—close to one-half million dollars per test per animal species. A gene mutation approach such as using Ames' Salmonella is frequently supplemented by DNA-damage testing and eukaryote-based testing. Nearly all such assays now give both false positive and false negative tests, but they are relatively inexpensive screening procedures which have proven correct as much as 80-90% of the time. Always positive and negative controls are carried out with the test chemical.



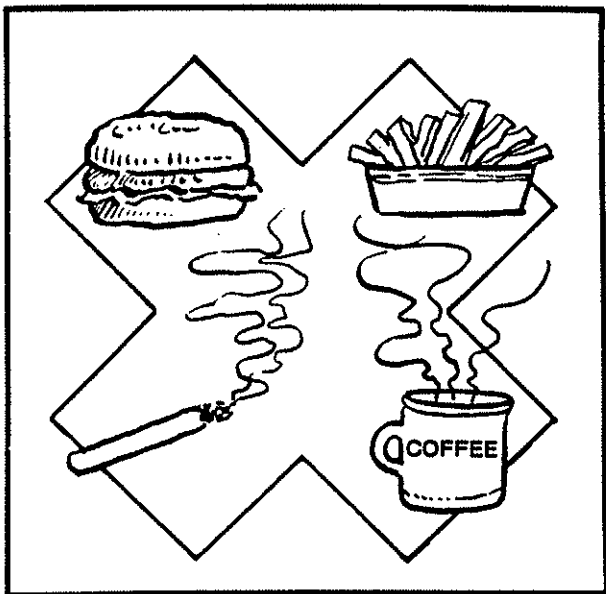
Animal Testing—The best method for assessing toxicity is by actual animal testing. There is a complete battery of tests that can be used to determine the toxicity of any substance. Because of the ancillary chemicals and chemical equipment required, the test animals involved, and the time expended, this is the most expensive method of assessing toxicity. Carcinogenesis testing in rats may follow the flow scheme below.

**A PLAN FOR DETERMINING THE CHRONIC RESPONSE OF AN ANIMAL
TO A TOXIC CHEMICAL**



Cost = \$300,000.00; 15,000 tissue samples examined; over 3,000 organs examined; 20,000 tests performed.



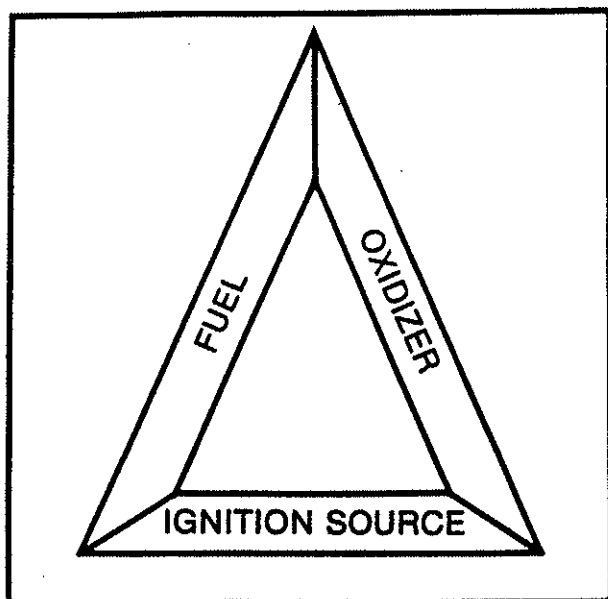


What Are the Proper Preventative Measures?

1. Unless you know definitely that a substance is not toxic, treat it as though it is.
2. Cover exposed areas of skin (wear gloves, aprons, caps, etc.).
3. Wear goggles.
4. Use a respirator or gas mask, as appropriate.
5. Use care in handling contaminated glassware and metal (USE CARE ALL THE TIME).
6. Change protective garments as required.
7. Keep food and cigarettes out of the lab.
8. Don't use lab glassware for eating or drinking.
9. Keep antidotes for common toxicants on hand.
10. Alert medical department as to what toxicants are being used, so they can be prepared.
11. Get periodic physical examinations—they uncover hidden problems.
12. Train all personnel thoroughly. Be sure they learn the toxicology of their work, where to find data, how to store and handle each substance, and what to do in the event of toxic poisoning.
13. OSHA details handling of certain carcinogens in CFR 29 1910.1000 et seq.



8. Flammables

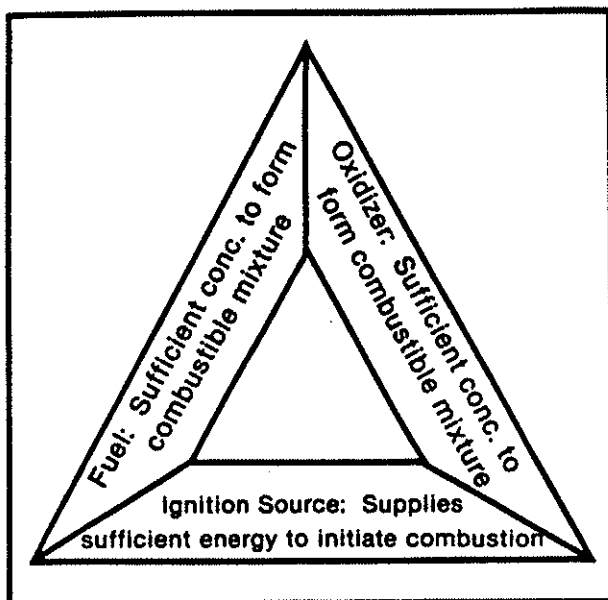


What is Burning?

Burning is the rapid oxidation of a fuel by an oxidizer (usually air) with the liberation of heat and (usually) light. The fuel is usually present in the gaseous state, though finely divided liquids and solids can burn.

How Does a Fire Start?

A fire can only be started when sufficient energy is present to initiate and sustain the chemical reaction which we call burning. Likewise, the fuel and oxidizer must be present in the proper proportions.



The process of burning involves three interrelated components: fuel, oxidizer, and ignition source. These three components make up the **FIRE TRIANGLE**. For a fire to occur, all three sides of the triangle must be present.

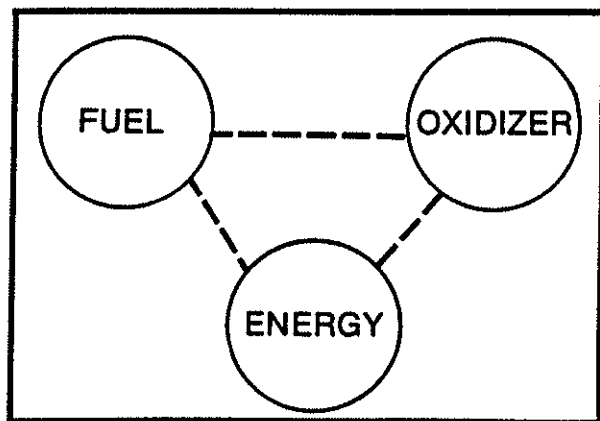
For practical purposes, the fuel (any substance capable of burning) and the oxidizer (usually air) must be present in sufficient quantity to form an ignitable mixture. To complete the triangle, the ignition source, containing sufficient energy to initiate combustion, must be present. The ignition source need not be in the form of a flame or spark; heat alone can supply the energy.



Typical examples of these three interrelated components are listed in the following table:

Examples of Necessary Components for Initiating Fire		
Fuels	Oxidizers	Ignition Sources
Flammable Liquids Acetone Alcohol Hexane Ether Toluene Pentane Flammable Solids Plastics Fibers Wood/Paper Dusts Pyrophoric Metals Flammable Gases Acetylene Propane Carbon Monoxide Hydrogen Ethylene Oxide	Gaseous Oxygen Fluorine Nitrous Oxide Oxygen Difluoride Ozone Chlorine? Liquids Hydrogen Peroxide Nitric Acid Perchloric Acid Bromine Solids Metallic Perchlorates Metallic Peroxides Ammonium Nitrate Ammonium Nitrite	Sparks Flames Static Electricity Heat Electrical Equipment Cigarettes Light Bulbs Heaters Lightning Catalytic Surfaces Lasers Self Heating

Source of heat
or temperature

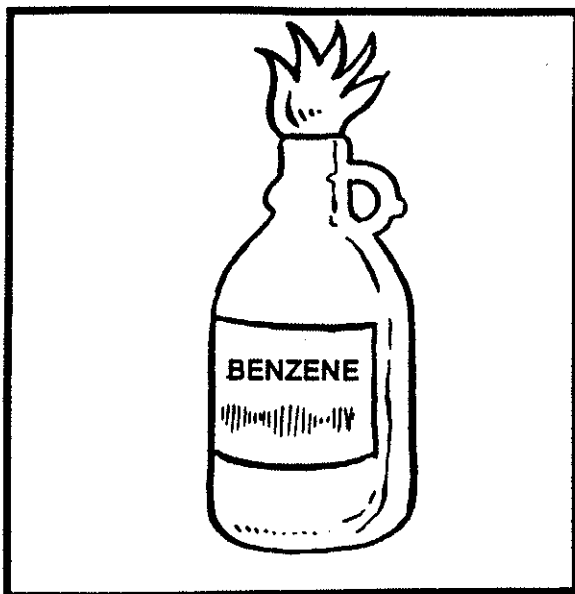


Removal of at least one side of the FIRE TRIANGLE or stopping the chemical reaction is the basis of fire control and flammable materials safety.

a) pyrophores:

phosphorus $\Rightarrow$ spontaneously
combust $> 88^{\circ}\text{F}$

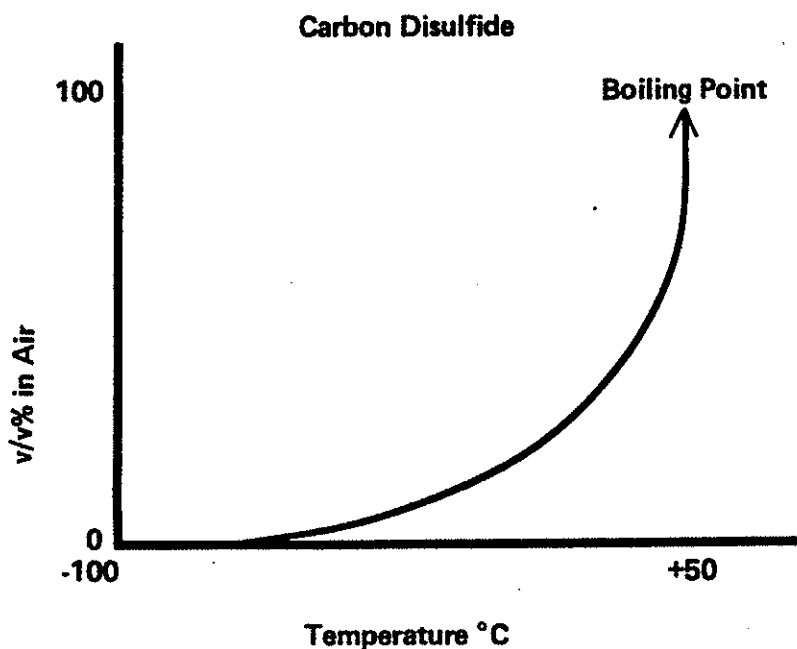


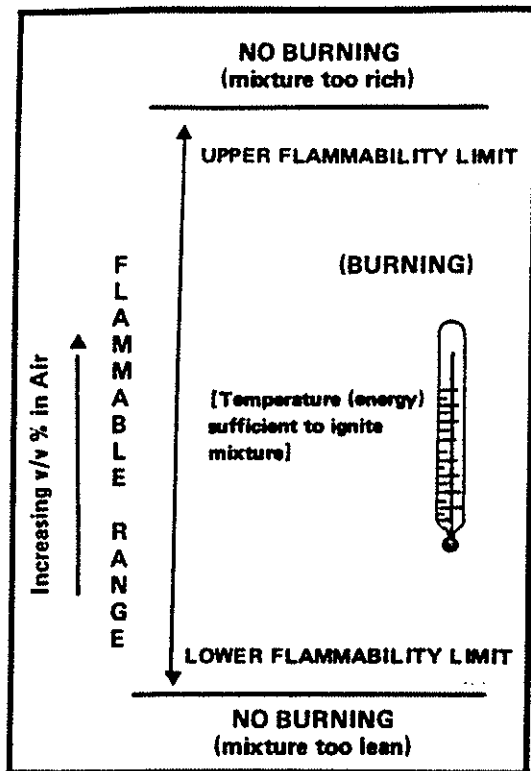


Flammable Solvents are the most common source of fires in the industry.

A flammable solvent is an organic liquid whose vapor can form an ignitable mixture with air. The solvent vapor is the fuel. The oxidizer is the surrounding atmosphere. For the mixture to burn, an ignition source must be present.

As temperatures are raised, the equilibrium vapor pressure of flammable solvents rise; i.e., the amount of solvent, v/v%, in contact with the atmosphere increases until it reaches 100%, the boiling point.

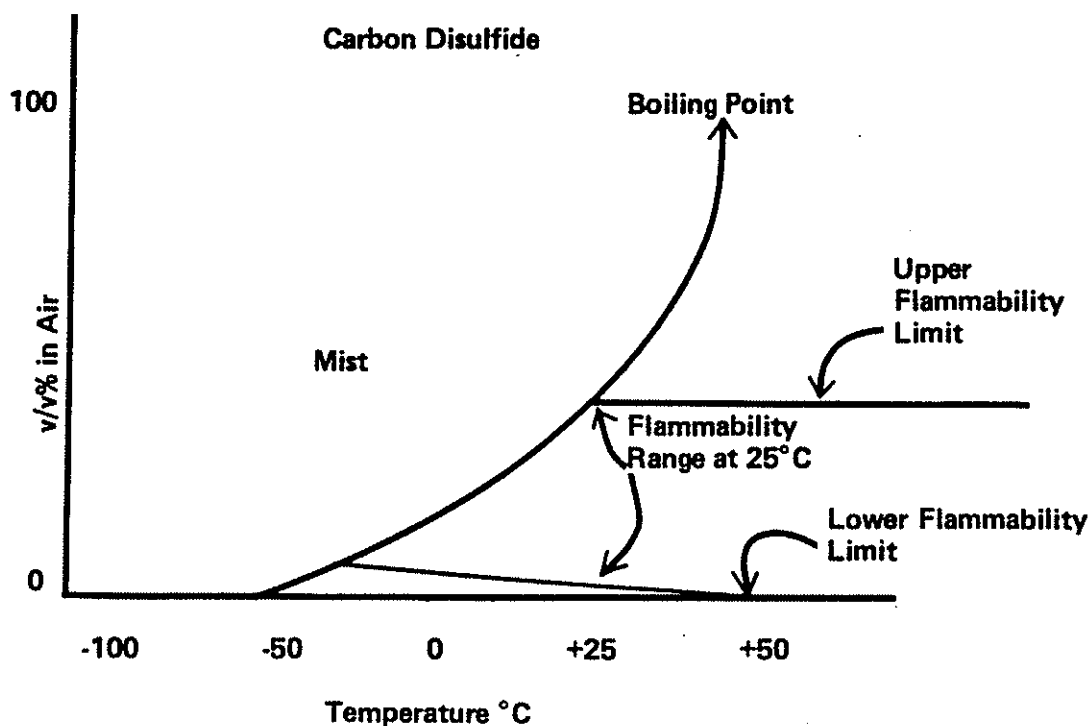




Flammability Limits are determined for flammable solvents by using a standard Bureau of Mines test apparatus. Homogeneous vapor-air mixtures are placed in a tube and subjected to an ignition source. The extremes in concentration between which the mixture burns are called the flammability limits.

Flammability limits are affected by a wide variety of physical conditions. In general, the range becomes larger for:

1. Upward flame propagation
2. Higher temperatures
3. Drier atmospheres
4. Higher total pressures
5. Greater ignition energies
6. Mist
7. Increased oxygen concentrations

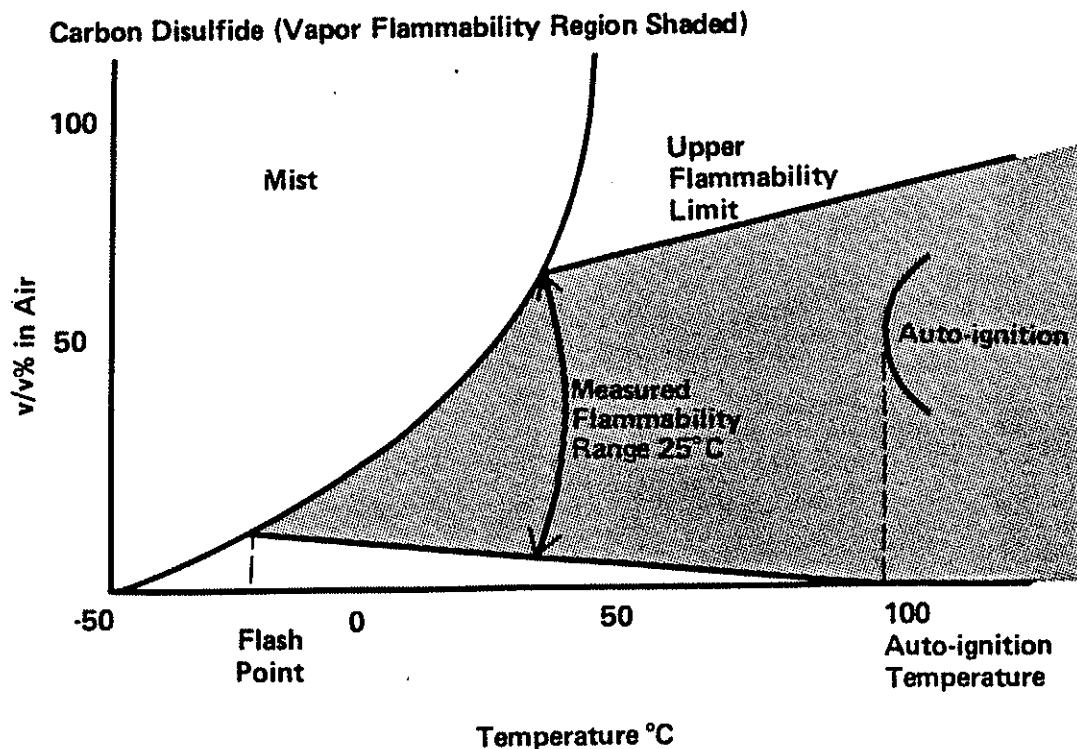


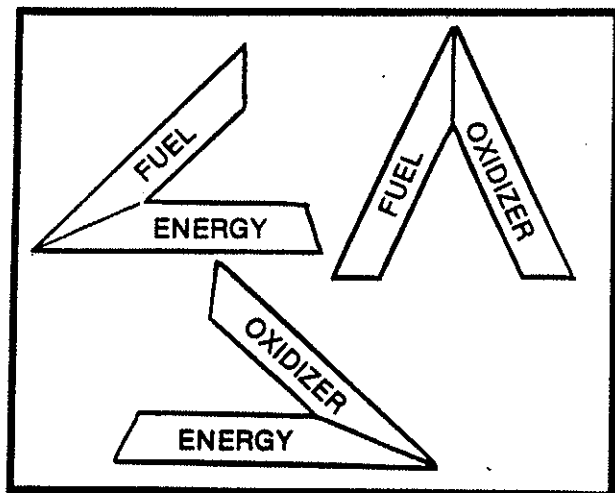
Flash Point Measuring Methods
Tagliabue Closed Cup
Tagliabue Open Cup
Pensky-Martens Closed Cup
Cleveland Open Cup
Setatester

Ignition Temperature (Auto-ignition Temperature)—The minimum temperature at which a substance will ignite in the air when there is no ignition source. For liquids it is defined as the lowest temperature at which a solvent will ignite spontaneously (measured as the temperature of a hot plate on which solvent ignites when added in drops). The auto-ignition temperature of toluene is 552°C and 100°C for carbon disulfide.

Flash Point—Is the lowest temperature at which a fuel-air mixture present above the surface of a liquid will ignite if an ignition source is introduced.

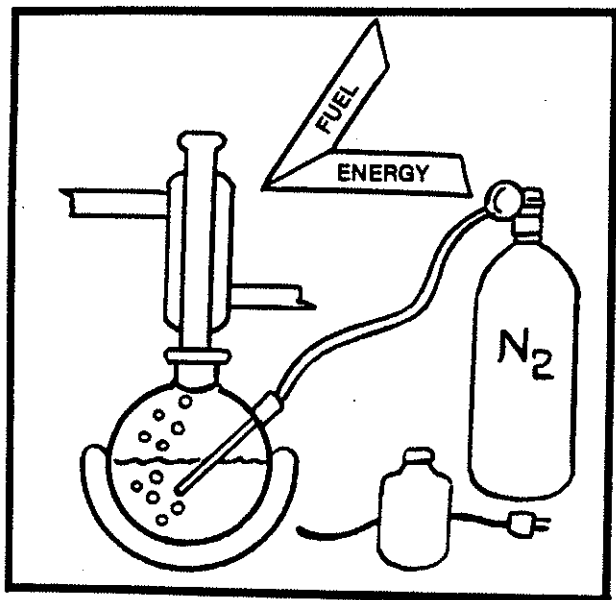
Fire Point—The lowest temperature within the flammable range that will allow propagation of flame across the surface of a liquid, away from the ignition source. Usually higher than the flash point, but may equal the flash point as in the case of carbon disulfide.





Control of Flammable Solvent Hazards

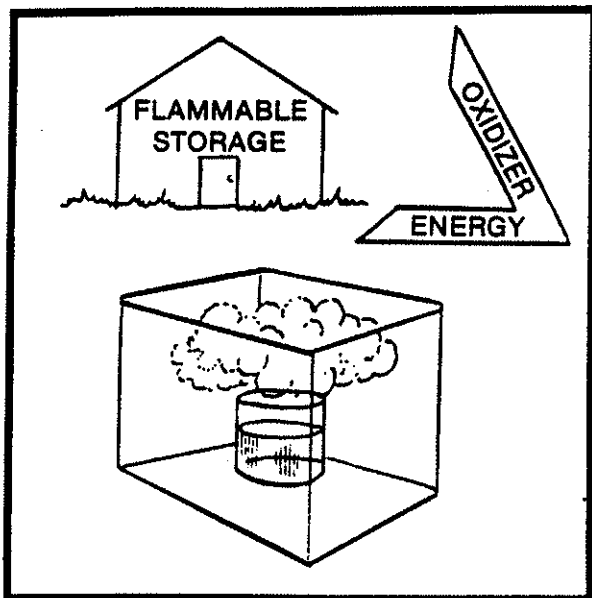
The control of the fire hazards of flammable solvents involves removing, or otherwise making inaccessible, at least one side of the FIRE TRIANGLE or blocking the chemical reaction itself. Any operation utilizing flammable solvents should be carried out outside the flammability region.



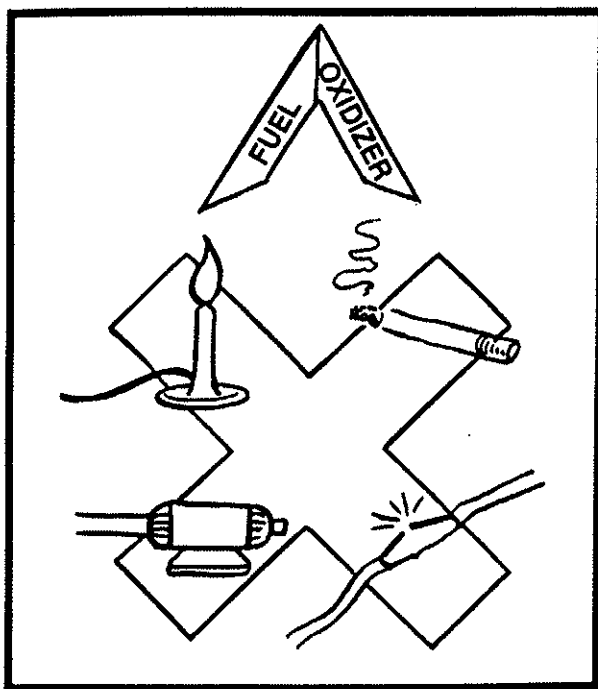
Preventative Measures

Removing the oxidizer is usually the most difficult approach. For some reactions involving flammable solvents, an overlayer of inert gas, such as nitrogen or carbon dioxide may be used to prevent fires. Carbon disulfide is non-flammable in oxygen-nitrogen atmospheres containing less than 5.3% v/v oxygen. If carbon dioxide is used to dilute air, carbon disulfide is non-flammable at oxygen concentrations below 7.6% v/v. Flammable solvents should always be isolated from oxidizers such as perchlorates, nitrates, peroxides, etc. See Chapter 16—Storage.





Removing the fuel is a somewhat easier approach. Store flammable solvents in areas which are well-ventilated to prevent build-up of vapors. When using or dispensing flammable solvents, use a working fume hood with a face velocity of at least 100 ft/min. When possible, work with flammables 10-12°C below the flash point. Do not mix solvents in waste cans or operate with many open solvent containers in one area. Keep cans, drums and other containers tightly closed. Use inhibitors known to decrease the vapor pressure of the solvent, i.e., water in acetone or alcohol, acetone in carbon disulfide.



Removing the ignition source is often the easiest approach to help eliminate hazards. It should be done routinely, but the other approaches should also be used simultaneously.

Mix halons (see Chapter 15) with the solvent or in the atmosphere above the solvent. Two to five percent halon can effectively render some flammables nonflammable.





Flammable Gases also exhibit the properties of the flammable solvents, but the dangers are magnified because boiling points are below room temperature. The gas is always present. Chapter 13 treats these problems in more detail.

The following pages list the properties of some commonly used flammable liquids and gases.



Fire-fighting (See Chapter 15)

Your local fire department can advise you of the equipment needed for the types of fire you might encounter. Generally, the obvious approach to controlling and extinguishing a fire is by removing the oxidizer leg of the triangle (once the fire is going, enough energy of reaction is being supplied to sustain it). Water, foam, and carbon dioxide are the usual methods used, the exact method being dependent on the nature of the fuel (solvent). The fire department should be advised of the hazardous chemicals used in your facility.



Properties of Some Common Flammable Liquids and Gases

Chemical Name	°F	Flash Point °C-CC	Auto-ignition Temperature °C	Boiling Point °C	Flammability Limits-25°C volume %	NFPA Rating
						<i>Hazard</i> <i>Flammability</i>
Acetaldehyde	-38	-38	175	21	4.0-60	2-4-2
Acetic Acid	103	40	465	118	4.0-20.0	2-2-1
Acetone	-4	-18	465	57	2.6-13	1-3-0
Acetylene		-15	305	-84	2.5-80	1-4-3
Acetonitrile	42	6		82	3.0-16	2-3-1
Ammonia			651	-2	15-28	3-1-0
Aniline	158	76	615	185	1.2**-8.3**	3-2-0
Benzene	12	-11	560	80	1.3-7.1	2-3-0
Carbon Disulfide	-22	-30	100	46	1.3-50	2-3-0
Carbon Monoxide			609	-191	12.5-74	2-4-0
Cresol (mixed)	201	88	559	195	1.3-	3-2-0
Cumene	96	44	424	153	0.9-6.5	2-3-0
Cyclohexane	-4	-17	245	82	1.3-7.8	1-3-0
Decalin	136	58	250	195	0.74*-4.9*	
Diethyl Ether	-49	-45	160	35	1.9-36	2-4-0
N,N-Dimethylformamide	136	57	435	153	2.2*-15*	1-2-0
Ethyl Acetate	24	-4	427	77	2.2-11	1-3-0
Ethyl Alcohol	55	13	365	88	3.3-19	0-3-0
Ethylene			490	-104	2.7-36	1-4-2
Ethylenimine		-11	320	56	3.6-46	3-3-3
Ethylene Oxide		-18	429	10	3.6-100	2-4-3
Formaldehyde			275	-19	7-73	2-4-0
Gasoline	-45	-43	280-456	38-204	1.4-7.6	1-3-0
n-Hexane	-7	-26	225	69	1.2-7.4	1-3-0
Hydrazine	100	38	270	113	4.7-100	3-3-2
Hydrogen			585	-252	4.0-75	0-4-0
Methyl Alcohol	52	12	430	64	6.7-36	1-3-0
Methyl Ethyl Ketone	16	-5	516	80	1.7-11.4	1-3-0
Methylene Chloride			615	40	16-66	2-0-0

(100% O₂)

(continued)

Properties of Some Common Flammable Liquids and Gases

Chemical Name	°F	Flash Point °C-CC	Auto-ignition Temperature °C	Boiling Point °C	Flammability Limits-25°C volume %	NFPA Rating
n-Pentane	-40	-40	260	36	1.4-7.8	1-4-0
Phenol	175	80	715	181	1.5	3-2-0
Propane			468	-44	2.1-9.5	1-4-0
Propyl Alcohol	74	25	440	97	2.1-14	1-3-0
Propylene Oxide	-35	-37		35	2.8-37	2-4-2
Pyridine	68	20	482	115	1.8-12.4	2-3-0
Styrene (monomer)	90	32	490	146	1.1-6.1	2-3-2
Tetrahydrofuran	6	-14	321	66	2.0-12	2-3-1
Tetralin	160	71	385	207	0.84*-5.0*	1-2-0
Toluene	40	7	480	111	1.2*-7.1	2-3-0
1,1,2-Trichloroethane			500	114		2-1-0
Xylene (mixed)	81-90	25	530	140	1.1*-7.0	2-3-0

*Measurement made at 100°C

**Measurement made at 140°C

All Flash Point Data Closed Cup

NFPA Ratings are for the 704M System described in Chapter 7. The numbers represent health, fire, and reactivity hazards, respectively. Higher numbers indicate increased danger in each category. Maximum rating is 4. Note that methylene chloride is given a flammability rating of 0 and will burn; ammonia is classified as a non-flammable by DOT, but will burn.



Common Safety Problems Relating to Flammable Solvents

1. Improperly stored glass containers of flammable solvents. (See Chapter 16)
2. A lack of or inadequate grounding to prevent static sparks, generated by the flowing liquids.
3. Mixed solvents in a waste disposal can. (See Chapter 12)
4. A lack of provision for solvent spill cleanup.
5. No floor level ventilation.
6. Smoking in solvent areas, ignition sources present.
7. Solvents, such as ether, stored in refrigerators which are not explosion proof.
8. Inadequate fire extinguishers or automatic extinguisher system.
9. Unsafe transport of flammable liquids (glass containers on carts carried in elevators, etc.)
10. Flammable solvents stored with oxidizers or other reactive chemicals.
11. Lack of adequate flame arrestors and explosion vents.

Common Control Measures

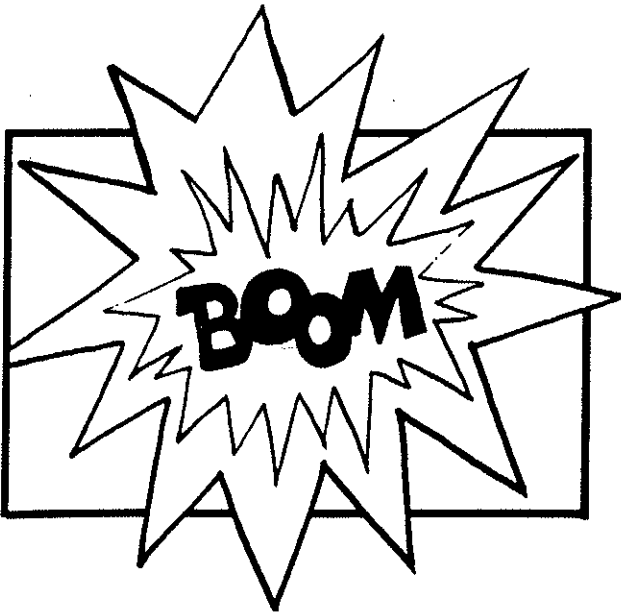
1. Isolate solvents from reactive chemicals.
2. DO NOT store more than a total of five gallons of solvents in glass in a laboratory. (See Chapter 16)
3. Use an inert atmosphere, if possible, when using solvents.
4. Transfer solvents in a working fume hood.
5. Eliminate sources of ignition.
6. Store solvents in a cool area, maximum temperature of 80°F.
7. Transport solvents in metal or other protective containers.
8. Ground all drums and transfer vessels when handling solvents. (Grounding clips must contact bare metal)
9. Thoroughly acquaint workers with the hazards of flammable solvents.
10. Have fire-fighting equipment close at hand AND KNOW HOW TO USE IT.
11. Work with and transport solvents only in a well-ventilated area.





Flammable Solids can form explosive mixtures in the atmosphere and react in much the same way as flammable gases. Such dangers are especially evident for particle sizes less than 5 μm .

Dust accumulates on surfaces and can be fanned into the air by a fire, air jet, or even an improperly chosen fire extinguisher. A layer of coal dust about one one-hundredth of an inch thick on the floor of a room 15' x 15' x 8' would fill the room with an explosive mixture if mixed with the air.



Removal of one arm of the fire triangle is the key to safety.



**Common Chemical Dust Problems
and Explosive Concentrations
in oz/1,000 ft³**

Aluminum	45
Allyl Alcohol Resin	35
Cellulose Acetate	25
Coal (low volatile)	125
Coal (high volatile)	55
Methyl Methacrylate	20
Magnesium	10
Polyamide	30
Polyethylene	20
Polystyrene	15
Tin	190
Titanium	45
Zinc	480

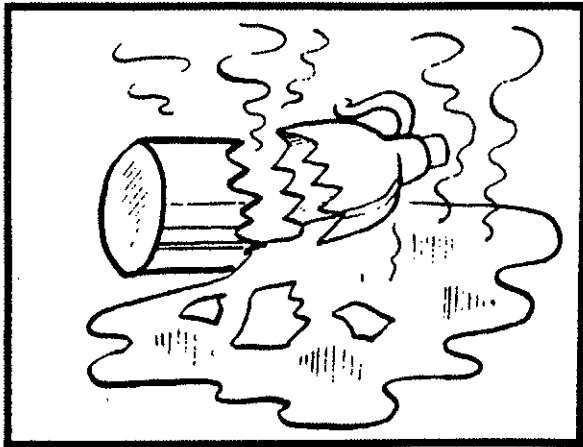
Dust explosion hazards are normally decreased by:

1. Larger particle size
2. Lower dust concentrations
3. Higher moisture
4. Additional inert material
5. Decreased oxygen concentration
6. Lower ignition energies and/or shorter ignition exposure times
7. Less porous dusts

Dangers can be minimized by:

1. Isolation from ignition sources
2. Frequent cleaning with vacuum or brush
3. Using dust-tight containers
4. Using adequate ventilation
5. Using external dust collectors
6. Introducing an inert gas
7. Constructing a facility to preclude dust buildup on walls, beams, etc.
8. Using proper fire extinguishers, arrestors, blowout walls, etc.





Flammable Solvent Spill Control

A solvent spill represents a potential fire hazard. Treatment methods should not only absorb the solvent, but should reduce the fire hazard as well. Therefore, a treatment agent that removes the fuel leg of the FIRE TRIANGLE should be the method of choice.

SOLUSORB® is an active, proprietary adsorbant, for just this purpose. In effect, **SOLUSORB** completely adsorbs the spilled solvent, and reduces the vapor pressure to such an extent that insufficient fuel (as vapor) is present to start a fire.



9. Corrosive Chemicals

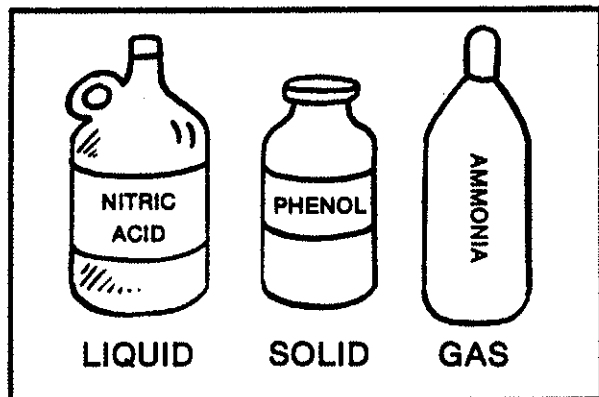


Corrosive chemicals are those substances that, by direct chemical action, are injurious to body tissue or corrosive to metal. Corrosive injury may be to a minor degree (irritation) or actual physical disruption of body tissues.

Corrosive chemicals act on body tissues through direct contact with the skin or eyes, inhalation, or ingestion.

Corrosive chemicals act directly on body tissues by:

1. Direct chemical reaction
2. Dissolution of essential components
3. Destruction of protein
4. Disruption of cell membranes



Corrosive chemicals will be discussed by considering the physical state of the material.

In this section, we will be concerned mainly with the practical aspects of corrosive chemicals and their hazards, rather than regulatory viewpoints.

What are the hazards?

How are they controlled?

How are hazard situations dealt with?



Corrosive Liquids

Liquid corrosives are, perhaps, the most important category in that they are involved in the most common type of corrosive injury, especially where external injury is concerned. Given below are typical corrosive liquids encountered in many workplaces.

Common Corrosive Liquids

Inorganic Acids

Chlorosulfonic
Chromic
Hydrochloric
Hydrofluoric
Nitric
Sulfuric

Organic Acids

Acetic
Butyric
Chloroacetic
Formic

Other Inorganics

Bromine
Phosphorous Trichloride
Silicon Tetrachloride
Sulfuryl Chloride
Thionyl Chloride
Peroxides

Caustic Solutions

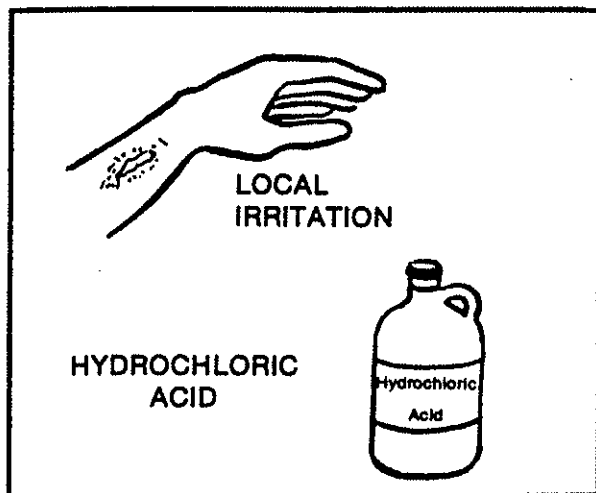
Ammonia
Sodium Hydroxide
Potassium Hydroxide

Organic Solvents

Dichloroethylene
Ethylene Chlorohydrin
Perchloroethylene
Methyl Ethyl Ketone
Gasoline

Other Organics

Acetic Anhydride
Liquified Phenol
Triethanolamine
2-Aminoethanol



The acids, caustics, other inorganics, 2-aminoethanol, and acetic anhydride are the compounds listed in the above table which require a DOT corrosive label. The others do not require such labeling, but are nonetheless destructive to tissue.

The primary modes of attack of corrosive liquids are the skin and eyes. There is a danger of corrosive vapors escaping from some solutions, i.e., ammonia, nitric acid, bromine and others. Such vapor attack will be considered later.



H₂SO₄
Sulfuric Acid

Physical Properties: Dense, oily corrosive liquid. Liberates heat when mixed with water. Dissolves most metals with release of hydrogen. Very strong dehydrating agent. Vapor pressure less than 0.001 mm Hg.

Flammability: Nonflammable, but highly reactive with finely divided combustible materials. Heat of reaction with metals can ignite hydrogen released.

TWA: 1 mg/m³

LD₅₀ (oral, rat): 2140 mg/kg

IDLH: 80 mg/m³

DOT: Corrosive

NFPA: 3-0-2-w

Local: Destruction of skin and eye tissue. Precipitates protein and dehydrates tissue.

Respiratory: Irritation of nose, throat, larynx, bronchia, lungs with destruction of tissue. Can induce bronchitis, pneumonitis, pulmonary edema.

Gastrointestinal: Dissolution of teeth, erosion, and destruction of tissue in mouth, throat, abdomen. Perforation of gastrointestinal tract.

Everyone knows mineral acids can cause burns. Few people realize, however, the extent to which they can damage body tissues. Corrosive action is related to toxicity as described in Chapter 3. Additionally, the concentration and duration of exposure controls the degree of injury. To exemplify the hazard situation, the corrosive and toxic effects of sulfuric acid, H₂SO₄, are highlighted at the left. Specific dangers of some other liquid corrosives are listed on the following pages.

NOTE: Sulfuric acid vapor is more toxic than hydrogen cyanide (TWA of HCN = 10 mg/m³).

Fuming sulfuric acid or oleum is concentrated sulfuric acid, 98% in water, with an excess of SO₃ dissolved in it. It is more reactive than sulfuric acid itself and much more hazardous.



HNO₃
Nitric Acid

Physical Properties: Colorless corrosive liquid which frequently is light red-brown because of its decomposition product, NO₂. Very reactive, strong oxidizer. Liberates heat when mixed with water. Vapor pressure as high as 100 mm Hg depending on wetness. Concentrated nitric acid is about 70% HNO₃ in water. Red fuming nitric acid has an excess of nitrogen dioxide dissolved in pure nitric acid.

Flammability: Nonflammable, but highly reactive with finely divided combustible materials. Heat of reaction with metals can ignite hydrogen released. Likely to cause ignition of wood and other cellulose products if finely divided. Acetic acid, aniline, acetone, nitrobenzene and some other organics react explosively with hot nitric acid and red fuming nitric acid.

TWA: 2 ppm NO₂

LC50 (inhalation rat):

65 ppm NO₂ four hours
244 ppm NO₂ thirty minutes

IDLH: 100 ppm

DOT: Corrosive

NFPA: 3-0-0-ox

Local: Destruction of skin and eye tissue by dehydration and oxidation. Oxidation of protein to yellow-brown xanthoproteic acid is typical.

Respiratory: Irritation of nose, throat, larynx, bronchia, lungs with destruction of tissue. Can induce bronchitis, pneumonitis, pulmonary edema.

Gastrointestinal: Dissolution of teeth, erosion, and destruction of tissue in mouth, throat, abdomen. Perforation of gastrointestinal tract.

HF
Hydrofluoric Acid

Physical Properties: A corrosive solution of gaseous hydrogen fluoride dissolved in water. Normally 48-60%, but may approach 100% in some applications. Dissolves glass but not polyethylene. The reaction with glass generates very corrosive SiF₄ gas. Very high vapor pressure approaching 760 mm Hg as the concentration approaches 100%.

Flammability: Nonflammable and normally neither a sufficiently strong acid or oxidizer to cause spontaneous ignition of organics or gaseous products from its reaction with metals.

TWA: 3 ppm

LC50: Not available, but animals die when exposed to 1500 mg/m³ for five minutes, while 100 mg/m³ could be tolerated for five hours without causing death.

IDLH: 20 ppm

DOT: Corrosive

NFPA: 4-0-0

Local: Causes more severe burns than other mineral acids. Penetrates the skin quickly. Instantaneous painful burns from contacting solutions of concentrations greater than 60%. Initial contact with more dilute solutions, even those less than 5%, may not cause a pain sensation, but skin whitening occurs with destruction of underlying tissue. Painful ulceration may follow.

Respiratory: Similar to nitric, but a systemic poison and may not be initially painful.

Gastrointestinal: Similar to nitric.



NaOH
Sodium Hydroxide Solution

Physical Properties: An oily corrosive solution of solid sodium hydroxide normally supplied as a 45-75% solution.

Flammability: Sodium hydroxide solutions are nonflammable.

TWA: 2 mg NaOH/m³

LD50: Not available, but a 1% solution causes severe damage to the eyes of rabbits.

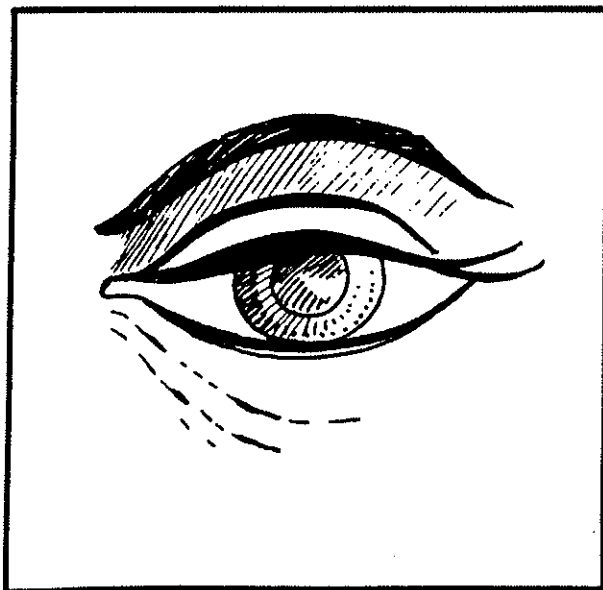
IDLH: 200 mg NaOH/m³

DOT: Corrosive

NFPA: 3-0-1

Local: See discussion to the right.

Concentrated alkaline solutions have a more corrosive effect on tissue than most strong inorganic acids. Strong alkalies gelatinize tissue, forming soluble compounds, thereby producing deep painful penetrating burns. For instance, tissue may become albumates and fats soaps. Initial contact may not be painful. Even relatively dilute solutions of sodium hydroxide, potassium hydroxide, ammonia, calcium hydroxide, sodium phosphate, and sodium carbonate, among others, can dissolve skin fats, soften the epidermis and sensitize the skin to other chemicals.



In the eye these caustics are extremely destructive and more difficult to remove than acids.

In fact, isotonic solutions of NaOH, 2%, cause irreparable damage to the eyes of rabbits within three minutes. Concentrated solutions, 50% or greater, cause irreparable damage within fifteen seconds.





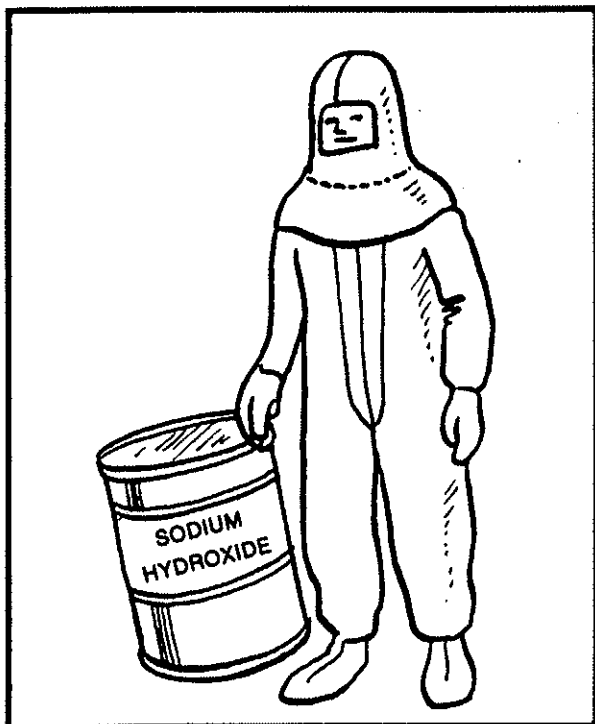
Precautionary Measures

Always wear adequate protective equipment (gloves, goggles, face shield, etc.). If the danger of inhaling corrosive fumes exists, wear suitable respiratory protection; immediate first-aid must include flushing with large amounts of water for 15 minutes followed by medical attention. Do not rub the skin.

Check the product label.

In case the chemical gets into the eye, it is absolutely imperative that the eyes are washed immediately with water. Seconds count! Occasionally the eye lids should be lifted during irrigation. Tests have shown that some chemicals need to be flushed for 15 minutes before leaving the area to seek medical attention. **CONTACT LENSES SHOULD NEVER BE WORN AROUND CORROSIVES!**

Do not induce vomiting if corrosives are ingested; instead have your physician suggest proper first-aid treatment.



For bulk handling, a "Chempruf suit" or equivalent is advisable.



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Alkalies

Calcium Oxide
 Calcium Hydroxide
 Cement
 Potassium Hydroxide
 Sodium Hydroxide
 Sodium Peroxide
 Sodium Phosphate

Organic Acids

Oxalic
 Phenol
 Salicylic
 Trichloroacetic

Metal Salts

Tin(IV) Chloride
 Phosphorus Pentoxide
 Potassium Chromate
 Mercury(II) Chloride

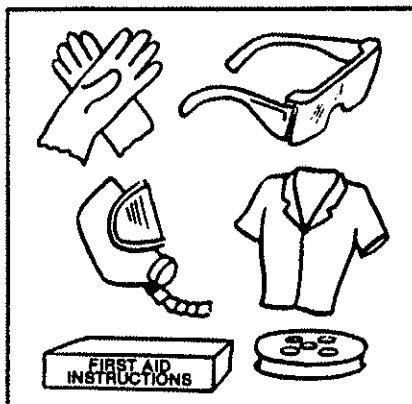
Elements

Sodium
 Potassium
 Lithium
 Phosphorus
 Iodine

Corrosive Solids

The effects of solid corrosives are largely dependent on their solubility in skin and respiratory moisture and, of course, on the duration of contact. Some compounds do have a specific ability to penetrate the skin even though their solubility in water is relatively low. Of these phenol and salicylic acid are mentioned in the table to the left. Phenol is a very weak organic acid and salicylic is stronger. Besides the intrinsic corrosive properties of solids, the heat of solution is often an important factor in damaging tissue.

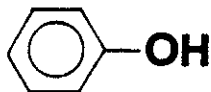
Dusting of corrosive solids presents the greatest danger in the industrial setting. Caustic alkalies, because of their frequent use, present the greatest hazard.

**Precautionary Measures**

As with liquids, adequate protective equipment (including respiratory) should be worn. In the finely divided state, many solid corrosives may be present in air and dust control is necessary. Immediate first-aid must include flushing with large amounts of water and securing immediate medical attention.

Full eye protection is particularly important.





Phenol*

Physical Properties: White solid, adsorbs water from air and liquifies. Soluble in water and most solvents.

Flammability: Combustible solid (FP 175° F, CC) evolves flammable vapors that can form explosive mixtures with air.

TLV: 5 ppm (skin)

LD50 (skin-rat): 669 mg/kg

Effects:

Local: Burns of skin and eyes.

Gastrointestinal: Burns of mouth, larynx, gastrointestinal tract, nausea, pain.

Respiratory: Breathing difficulty, cough, cyanosis, pulmonary edema.

Central Nervous System: Headache, dizziness, visual impairment, convulsions, unconsciousness.

Hazard

Besides being corrosive, it is easily absorbed into system and can be fatal.

Combustible solid.

Relatively more toxic than HCN (10 ppm).

The effects listed may be acute or chronic, local or systemic.

Prolonged contact with phenol can produce gangrene.

*Synonyms: Carbollic acid, hydroxybenzene, oxybenzene, phenic acid, phenyl hydrate, phenyl hydroxide, phenylic alcohol; fenol (Dutch, Polish), Fenolo (Italian)



Group I Very Soluble, Upper Respiratory Tract, or Very Irritating Ammonia Hydrogen Chloride Hydrogen Fluoride Formaldehyde Sulfonyl Chloride Thionyl Chloride
Group II Soluble, Upper Respiratory Tract and Bronchi Arsenic Trichloride Bromine Chlorine Iodine Phosphoric Pentachloride Phosphorous Trichloride Sulfur Dioxide
Group III Least Soluble, Easily Reach Alveoli Carbonyl Chloride (phosgene) Nitrogen Dioxide Ozone

Corrosive Gases

The most serious hazard associated with corrosives is from materials in the gaseous state. In this state, corrosives are readily absorbed into the body by dissolution in skin moisture, and by inhalation. Gaseous corrosives are usually grouped by solubility and the effect on the respiratory system. Typical examples are given at the left.

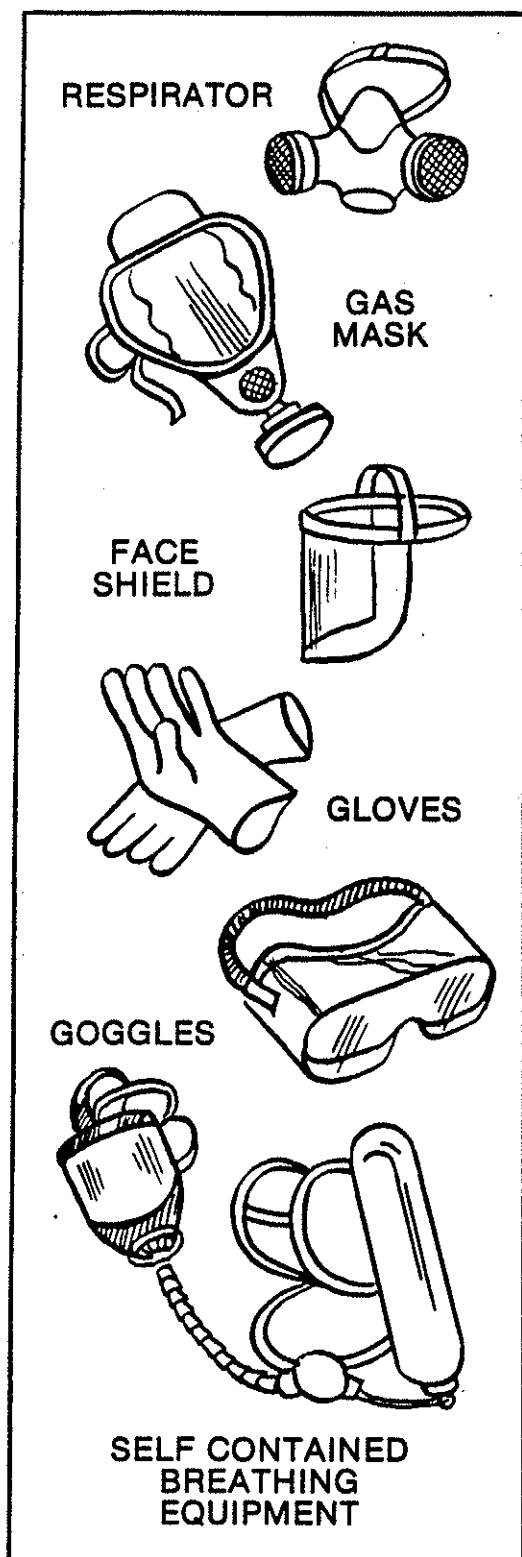
Group I compounds usually cause severe nose and throat irritations and, because of their high water solubility, do not penetrate deeply into the respiratory tract during the initial moments of exposure.

Group II compounds attack the upper respiratory tract and bronchi during initial exposure.

Group III compounds reach the alveoli quickly, and frequently there is no nose or throat irritation to warn of the presence of the gas. Delayed pulmonary edema may result.

A further group of toxic gases which asphyxiate or do systemic damage are considered in Chapter 3. Those discussed here are primarily corrosive to tissue.





Corrosive Gases (cont.)

Precautionary Measures

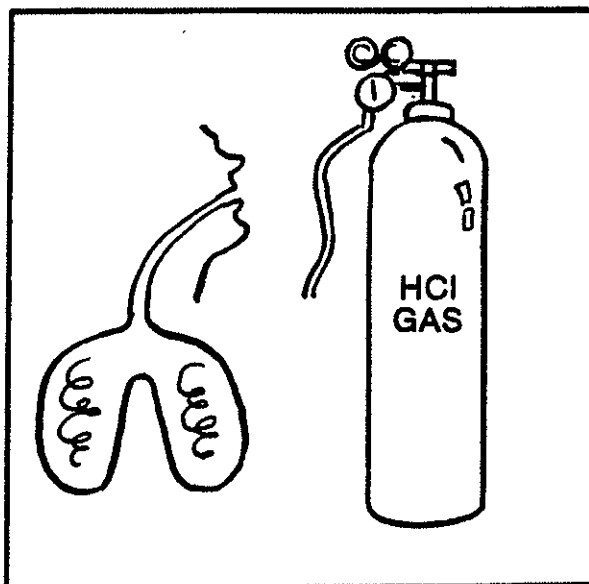
Of primary importance is the use of adequate respiratory protection, usually a self-contained breathing apparatus. Protective equipment for skin and eyes, of course, is necessary. A good exhaust ventilation system is highly recommended for those areas where corrosive gases may be encountered. Immediate medical attention should be secured on exposure.

All corrosive gases and vapors from corrosive liquids or solids also cause severe eye irritation or damage. It is absolutely imperative that contact lenses not be worn when working with these materials.

Skin and eyes contacted by corrosive gases should be washed thoroughly as suggested for liquids.

Escape breathing apparatus and escape lines are necessary in areas where corrosive gases might be encountered. Workers in such areas need to be trained in artificial respiration and CPR, and should plan escape routes.





Hazards

The harmful effect of a gaseous corrosive is not a linear relationship between concentration and exposure duration. To compound the problem, there are PRIMARY effects that can produce severe, immediate damage, and even death without causing systemic injuries, and vice-versa. The concentration, solubility, and duration of exposure must be considered when evaluating the possible effects.

Below, two common gaseous corrosives, ammonia and phosgene, are compared. Notice that phosgene, though less severe in local immediate effect, has severe long-range systemic effects.

Notice that phosgene, a Group III compound, cannot be detected by either smell or irritation below the TLV® but ammonia, a Group I compound, is easily detected. This exemplifies the typical characteristics of these groups.

<u>AMMONIA</u>	Physical Properties	<u>PHOSGENE</u>
Colorless gas with a sharp irritating odor. Soluble in water and many solvents.	Flammability	Colorless gas which hydrolyzes in water to form HCl and CO ₂ . Soluble in some solvents.
Nonflammable according to DOT, but can burn violently under the right conditions. See Chapter 8.	TLV®	Nonflammable according to DOT tests, but it will burn.
25 ppm	IDLH	0.1 ppm
500 ppm	LC50 (rat)	2 ppm
4837 ppm/1 hr	Nose Irritation	110 ppm/0.5 hr
20 ppm	Effects	10 ppm
Immediate burning, respiratory edema.	Delayed respiratory edema.	



Control of Corrosive Hazards

The control of corrosive hazards includes the precautionary measures set down earlier. There are additional precautionary measures and handling and storage techniques that should be applied to ensure adequate safety in working with corrosive materials. The table below is meant to summarize them.

Personal Measures

1. Wear adequate protective equipment to control external exposure hazard.
2. Have available (and use) adequate respiratory protection (respirator, gas masks, self-contained breathing apparatus).
3. Provide adequate exhaust ventilation where corrosive hazards are used.
4. Provide easy access to eye-wash stations.
5. Provide easy access to safety showers.
6. Provide easy access to immediate medical attention.
7. Secure periodic medical checkups.
8. Train all personnel thoroughly in the hazards of the materials worked with and first-aid measures to be taken.

Measures for Materials Protection

1. Store in cool, dry, well-ventilated areas.
2. Store away from sunlight and rapid temperature changes.
3. Store away from reactive chemicals, organics, and other potential hazards, such as combustibles and toxicants.
4. Store corrosives only in approved containers.
5. Store only under approved conditions.
6. An automatic sprinkler system or other water-spray device should be available.
7. Adequate fire-fighting equipment should be available.
8. A means of neutralizing and cleaning up corrosive spills should be available.



Principles of Corrosive Chemical First-Aid

1. In the event of contact with the eyes, immediately flush the eyes with water and continue to flush for 15 minutes.

The first few seconds after contact are critical, and immediate flushing of the eyes may prevent permanent damage.

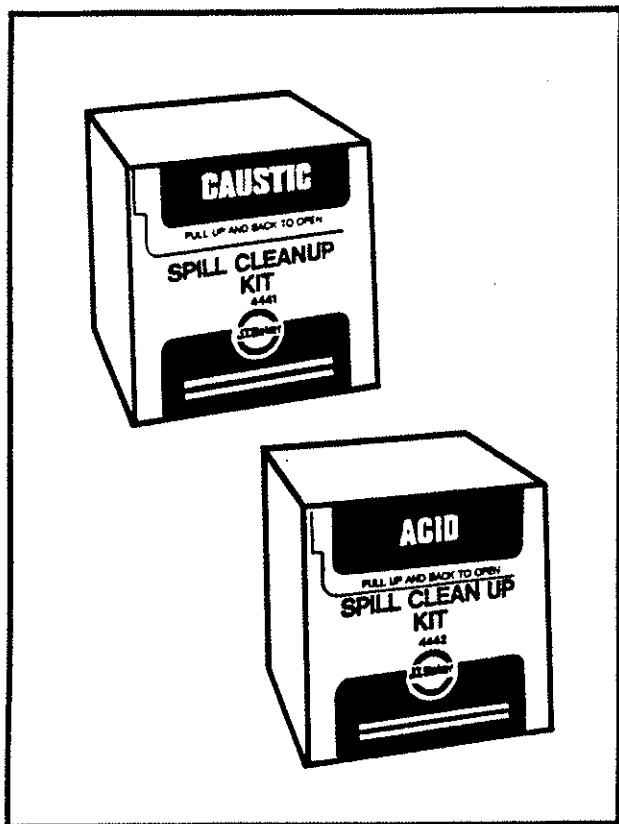
An eyewash fountain, an eyewash hose or any other source of water should be used in an emergency.

Remember, the one and only emergency treatment is to dilute the chemical immediately by complete flushing with water. The patient's eyelids may have to be forced open, so that the eyes may be flushed.

Alkali (base) burns are usually more serious than acid burns.

2. Strong chemicals burn the skin rapidly. There is no time to waste. Begin flushing the area with water immediately. Remove and discard clothing including socks and shoes (obtain other clothes and shoes). Continue to flood the area, while clothing is being removed. In the case of contact by sulfuric acid, remove clothes first.
3. The precautionary warning on the product label should be consulted for full first-aid information. Provide the label information to the attending physician. Obtain complete first-aid and emergency response data before handling the chemical.
4. Neutralizers and solvents (alcohol, etc.) should not be used by the first aider. The spread of a skin-absorbing corrosive poison, like phenol, can result in death.



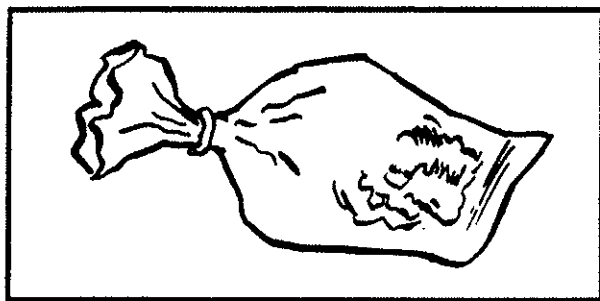
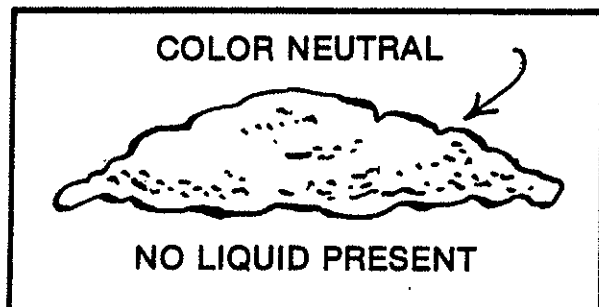


How To Deal With A Corrosive Chemical Spill

J. T. Baker has developed products for the safe cleanup of corrosive materials, such as mineral acids and strong caustics, for both laboratory (1-5 pints) and bulk (> 5 pints) spills. Both the Acid and Caustic Spill Cleanup Kits allow the rapid, safe cleanup of a variety of strong acids and caustics. All materials necessary are provided: treatment agents, gloves, safety glasses, brushes, scoops, sponges, disposal bags, and ties. The systems for both acidic and caustic spill neutralization are self-indicating, eliminating the need for guess-work, as to when the spill is neutralized.

The basic instructions for use are given on the next page.





Acids* and Caustics

Read the precautionary label information when using a hazardous chemical. Always take the necessary precautions. Respiratory protection is usually required.

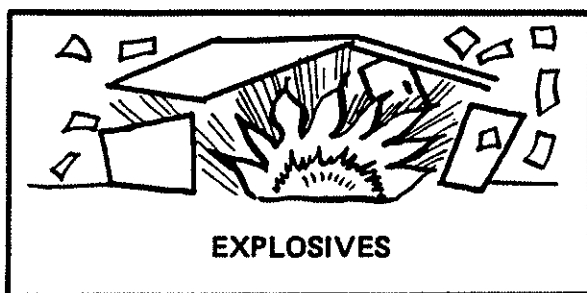
1. Apply NEUTRASORB® for acids and NEUTRACIT® 2 for caustics from the perimeter of the spill, inward. Use sufficient material to secure a uniform color change. When foaming subsides, proceed to the next step.
2. Carefully mix with the scoops provided. Carefully add water and more neutralizer if necessary.
3. Mix thoroughly until a color change occurs that is uniform throughout and indicates complete neutralization. Add the neutralizer/adsorbant as needed to effect complete neutralization.
4. Pick up neutralized spill with the scoops provided and transfer to disposal bag. Wipe up surfaces with provided sponge. Place used sponge, scoops and gloves in disposal bag, seal with tie and label.
5. Dispose of bag and contents in accordance with local regulations.

*Not intended for use with hydrofluoric acid.

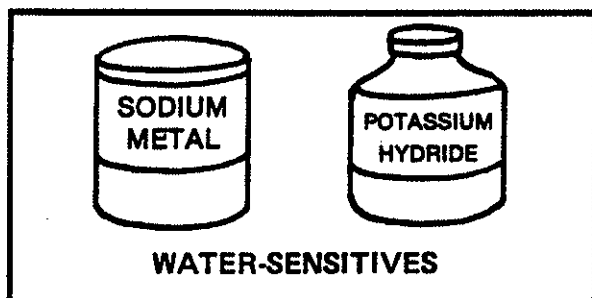


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11. Reactive Chemicals

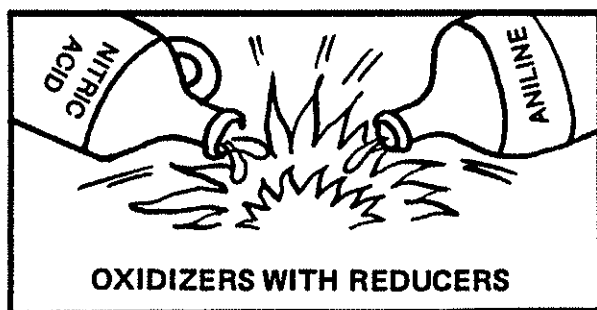
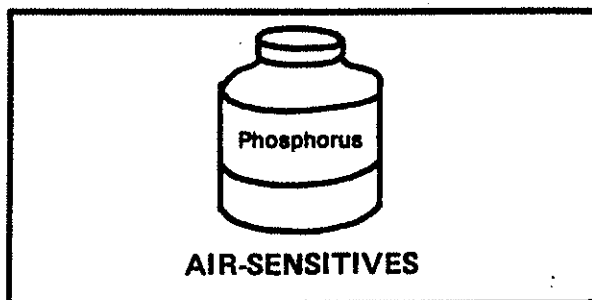


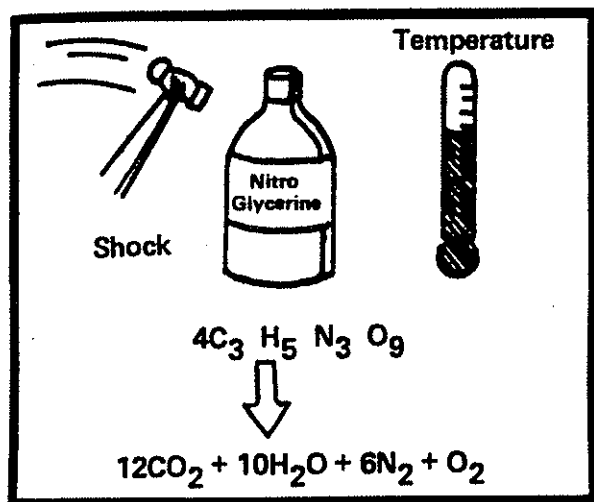
Reactive chemicals are substances which enter into violent reactions during which the spontaneous liberation of heat and/or gases is too rapid to be safely dissipated by the surroundings. Thus, the reaction is out of control and either the vessel bursts, an explosion occurs, toxic vapors are uncontrollably liberated, flammable gas is evolved, spontaneous ignition occurs, or any number of similar incidents occur such as those characterized in the drawings to the left.



We will categorize reactive chemicals in two groups:

1. Unstable Compounds
 - a. Explosives (e.g., TNT)
 - b. Others (e.g., monomers)
2. Unstable Mixtures
 - a. Explosives (purposefully mixed oxidizers and reducers, such as black powder)
 - b. Water-Reactive Compounds (e.g., sodium metal)
 - c. Air-Sensitive Compounds (e.g., phosphorus)
 - d. Other Mixtures of Oxidizing Agents with Reducing Agents (e.g., perchloric acid and cellulose)





Explosive Hazards

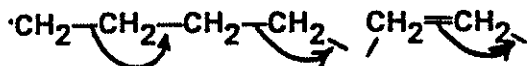
1. Possible physical destruction of life and property by shock front and associated noise, brisance.
2. Initiation of fire and other hazardous reactions—a chain reaction.
3. Rapid release of corrosive or toxic substances into the environment.

Polymerization

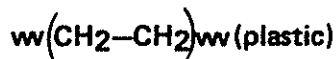
1. A monomer such as ethylene



2. Combines with another ethylene monomer and another, and another...



3. Until a long chain, the polymer is formed from thousands of ethylene molecules.



1. Unstable Compounds

a. Explosives

Explosives are substances which, under certain conditions of shock, temperature or chemical reaction, decompose with violent rapidity, usually releasing large quantities of gases, and heat. The process of explosion causes the rapid expansion of surrounding air, sometimes accompanied by burning gases and flying objects. This process is usually destructive to both life and property.

High explosives detonate at rates up to four miles per second, while low explosives detonate at rates of only thousands of feet per second.

DOT hazard classes for explosives are discussed in 49 CFR 173.53, 173.88, and 173.100.

b. Others

Many of the compounds listed on the next page may not always decompose explosively, but may still cause appreciable damage. Among these are the monomers.



COMMON EXPLOSIVE COMPOUNDS

1. Acetylenic Compounds

- a. Acetylene $\text{H}-\text{C}\equiv\text{C}-\text{H}$
- b. Copper(I) Acetylides $\text{Cu}-\text{C}\equiv\text{C}-\text{Cu}$
- c. Ethoxyacetylene $\text{C}_2\text{H}_5\text{O}-\text{C}\equiv\text{C}-\text{H}$
- d. Propyne $\text{H}-\text{C}\equiv\text{C}-\text{CH}_3$
- e. Propyne Peroxide $\text{H}-\text{C}\equiv\text{C}-\text{CH}_2-\text{OOH}$
- f. Disilver Acetylides $\text{Ag}-\text{C}\equiv\text{C}-\text{Ag}$

2. Azides

- a. Benzenesulphonyl Azide
- b. Carbonyl Diazone
- c. Silver Azide
- d. Lead(II) Azide

3. Azo Compounds

- a. Diazirine $\text{H}_2\text{CN}=\text{N}$
- b. Azomethane $\text{H}_3\text{C}-\text{N}=\text{N}-\text{CH}_3$
- c. Diazomethane CH_2N_2

4. Chlorite/Chlorate/Perchlorate Compounds

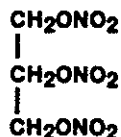
- a. Silver Chlorite AgClO_2
- b. Silver Chlorate AgClO_3
- c. Potassium Chlorite KClO_2
- d. Lead Tetrachloride PbCl_4

5. Fulminates

- a. Silver Fulminate AgCNO
- b. Mercury(II) Fulminate $\text{HgC}_2\text{N}_2\text{O}_2$
- c. Sodium Fulminate NaCNO
- d. Copper(II) Fulminate $\text{CuC}_2\text{N}_2\text{O}_2$

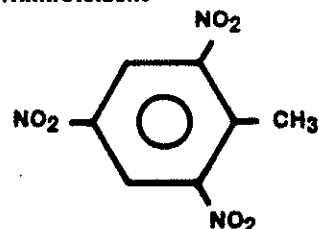
6. Nitro Compounds and Nitrate Esters

- a. Nitromethane $\text{H}_3\text{C}-\text{NO}_2$
- b. Nitroglycerine



6. Nitro Compounds (continued)

- c. Nitrocellulose
- d. Trinitrotoluene



7. Other Nitrogen-Containing Compounds (Diazonium Salts, Nitroso Compounds, Amides, Nitrides)

- a. Silver Amide AgNH_2
- b. Silver Nitride AgN_3
- c. Nitrosylcyanide $\text{O}=\text{N}-\text{C}\equiv\text{N}$
- d. Disulfur Dinitride $\text{S}=\text{NSS}-\text{NS}$

8. Picrates

- a. Picric Acid $\text{C}_6\text{H}_3\text{N}_3\text{O}_7$
- b. Lead Picrate $\text{Pb}(\text{C}_6\text{H}_2\text{N}_3\text{O}_7)_2$

9. Peroxides

- a. Diacetyl Peroxide $\text{C}_4\text{H}_6\text{O}_4$
- b. Zinc Peroxide ZnO_2
- c. Dimethyl Peroxide $\text{H}_3\text{C}-\text{OO}-\text{CH}_3$

10. Strained Ring Compounds

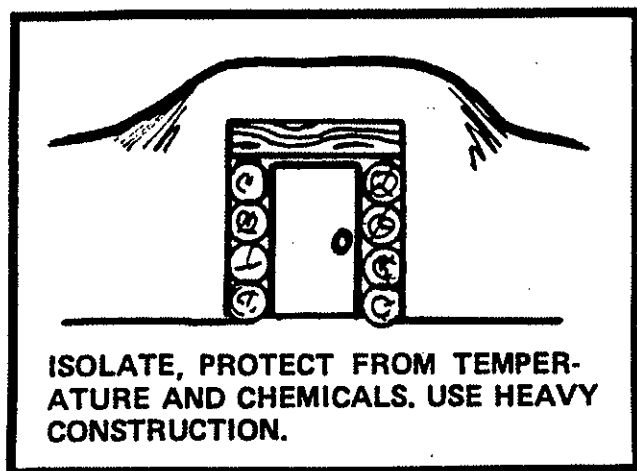
- a. Benzvalene C_6H_6
- b. Prismane C_6H_6

11. Polymerizable Compounds

- a. Acrylic Acid
- b. Ethylene
- c. Butadiene
- d. Styrene
- e. Vinyl Chloride
- f. Cyclopentadiene

See: *Handbook of Reactive Chemical Hazards*, L. Bretherick, Published by Butterworths.





Precautionary Measures for Unstable Compounds

In general, protect these materials from shock, elevated temperature, light, ignition sources, and other reactive chemicals.

Store a minimum quantity of the material and do not store in or near an area populated by workers.

Whenever possible, use the chemicals with added inhibitors.

Store explosives in isolated, heavily constructed magazines, preferably underground or at least in a building with a blow-out wall.

See Chapter 16 for more details.

Reactive Chemical	Inhibitor
Picric Acid	Water
Methyl Vinyl Ether	Triethanolamine
Cyclonite	Bees Wax
Lead Azide	Water
Lead Styphnate	Water
Acrylaldehyde	Hydroquinone
Disodium Acetylde	Water
Methyl Methacrylate	Hydroquinone
Vinyl Chloride	Phenol
Tetrafluoroethylene	Alpha Terpinene
Chlorotrifluoroethylene	Tributylamine



EXPLOSIVES
Nitroglycerine and Ammonium Nitrate Nitroglycerine and Potassium Nitrate

WATER REACTIVE COMPOUNDS
1. Alkali Metals (e.g., lithium, sodium) 2. Organometallic Compounds (e.g., tetramethylaluminum) 3. Halides (e.g., acetyl chloride, titanium tetrachloride) 4. Hydrides (e.g., diborane, sodium hydride) 5. Peroxides (e.g., sodium peroxide) 6. Carbides (e.g., calcium carbide) 7. Oxides (e.g., sodium oxide) 8. Phosphides (e.g., aluminum phosphide) 9. Anhydrides (e.g., acetic anhydride) 10. Other Hydrolyzable Compounds (e.g., chlorosulfonic acid, aluminum tribromide)

2. Unstable Mixtures

a. Explosives

Some examples of mixtures which are purposefully prepared as explosives are shown to the left. Precautions are the same as those listed on page 11-4.

b. Water-Reactive Chemicals

Some examples of water reactive compounds are listed to the left. All of the specific examples given have an NFPA rating which carries the special water-reactive symbol, ω .

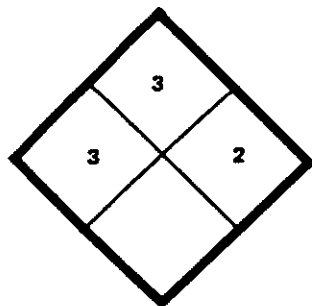
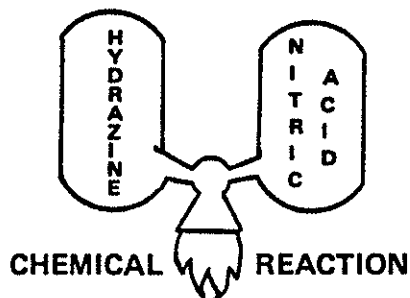
Compounds which are water reactive should not be stored where automatic sprinkler systems are installed. They should be properly desiccated whenever possible. Isolate from other reactive compounds. Store in a cool, water-proof area. No water should service the storage area. Use adequate protective equipment when handling. **KEEP AWAY FROM WATER.** Placard area with "Do not use water."



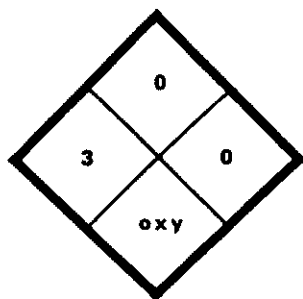
AIR-SENSITIVE COMPOUNDS

1. Metallic Dusts (e.g., zinc, nickel, titanium)
2. Alkali Metals (e.g., cesium, potassium)
3. Hydrides (e.g., diborane, barium hydride)
4. Others (e.g., diborane tetrachloride, dimethyl arsine, sodium amide, cyclopentadienyl sodium, and many organo-metallics)

MIXTURES OF STRONG OXIDIZERS WITH STRONG REDUCERS CAN BE USED FOR ROCKET FUELS



Hydrazine



Nitric Acid

c. Air-Sensitive Compounds

The materials listed to the left react spontaneously with oxygen in the air. All of them usually are pyrophoric as well. That is, they burst into flame on contact with the air.

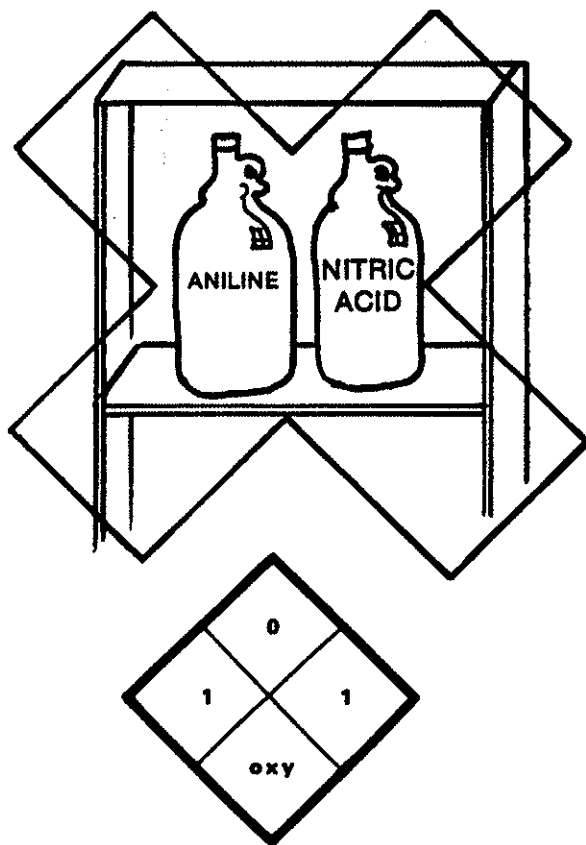
Pyrophoric metals such as finely divided nickel and titanium should normally be stored in metal containers with some moisture. Other solids are frequently stored under an inert gas or liquid.

d. Strong Oxidizing Agents and Strong Reducing Agents

Most of the unstable mixtures listed on the previous two pages are combinations of oxidizing and reducing agents, but represent conditions where special dangers exist (explosives, water reactive, and air reactive). The following page lists commonly encountered oxidizing and reducing agents. Mixtures of any one of the oxidizing agents with any one of the reducing agents is usually catastrophic. Moreover, each individual compound is capable of reacting violently with a wide variety of relatively inert compounds.



STRONG OXIDIZERS		STRONG REDUCERS
Fluorine Ozone Chlorine Persulfates Peroxides Peroxy Acids Perchlorates Dichromates Chromates Permanganates Hypochlorites Nitrates Nitrites	Nitrous Oxide Liquid Oxygen Liquid Air Chlorosulfonic Acid Nitromethane Hypochlorites Chlorates	Finely Divided Metals Hydrazine Hydrides Hydrogen Aniline Sodium Lithium Potassium Butadiene Acetylides



Ammonium Dichromate

Precautionary Measures

In general, isolate from other potentially reactive compounds. Use adequate protective gear when working with such agents. If the substances involved are not water-sensitive, safety showers, sprinkler systems, and flushing hoses should be made available. See the section, Storage and Handling of Hazardous Materials for further details.

All oxidizers in the table above carry the special NFPA symbol, OXY. The reducers carry no special warning.



Precautionary Measures

1. Isolate reactive chemicals. Store compounds of a class together, away from other classes.
2. Have plenty of water available for flushing where water-sensitive chemicals are not involved.
3. For water-sensitive chemicals, do not allow water in the storage area, or water-type fire extinguishers and sprinklers.
4. In general, store reactive chemicals in a cool, dry area, protect from sunlight, and label with outdate.
5. As a rule, protect reactive chemicals, especially explosives, from shock.
6. Store away from flammable and toxic materials.
7. Wear adequate personal protective equipment. If toxic gases may be generated, self-contained breathing apparatus is warranted.
8. Train workers thoroughly in the hazards of the chemicals they are working with.
9. Have antidotes, neutralizers, fire-fighting equipment, etc., at hand in case of emergency.
10. Have medical assistance available near those areas using reactive chemicals.

Storage and Handling Summary

See the section Storing and Handling Hazardous Chemicals and literature noted in your Safety Library for details on class groups and individual compounds. Given at the left is a summary of some of the points made in this section.



COMMON REACTIVE CHEMICALS

Ammonium Nitrate
 Ammonium Perchlorate
 Benzoyl Peroxide
 2-Butanone Peroxide
 tert-Butyl Hydroperoxide
 tert-Butyl Peroxide
 Calcium Hydride
 Cesium
 Chromium Nitrate
 Chromium Trioxide
 Diborane
 Diethylaluminum Hydride
 Dimethyl Phosphine
 m-Dinitrobenzene
 p-Dinitrobenzene
 Divinylbenzene
 Germane
 Hydrazine
 Hydrazine Hydrate
 Hydrazinium Chlorate
 Hydrazinium Chlorite
 Hydrogen Peroxide (concentrated)
 Hydroxylammonium Nitrate
 Lauroyl Peroxide
 Lithium Acetylde
 Magnesium Perchlorate
 Mercury(I) Perchlorate
 Nitroethane
 Nitromethane
 2-Nitropropane
 o-Nitrotoluene
 p-Nitrotoluene
 Peroxides¹
 Phosphorus (red)
 Picric Acid
 Potassium
 Trinitrotoluene

Violent fires and/or explosions are the hazards associated with the following reactive chemicals. They should be segregated in storage. Use this table to check your stock.

Check the tables in this chapter, page 12-5, and Chapter 16 for further details on dangerous chemicals which may react violently.

¹See page 12-5



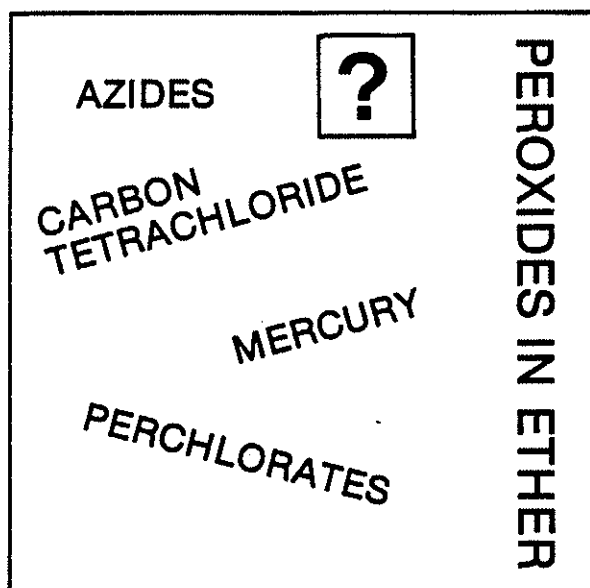
12. Insidious Hazards



What Are Insidious Hazards?

Insidious hazards are conditions within the laboratory that represent potential health hazards and, because they are not usually seen, smelled, tasted or felt, are easily overlooked and ignored. However, they may cause local or systemic, acute or chronic effects, dependent on the nature of the substance and the duration of exposure. In addition, insidious hazards represent a problem to workers in that they may never be aware that a problem exists until chronic, systemic poisoning has occurred.

This section details such hazards and their control.



All too often insidious or hidden hazards are overlooked during routine safety inspections.

Substances, such as mercury, present in microdroplets on a floor, can emit toxic vapor over long periods of time. Explosive perchlorates can form in fume hoods and ventilation systems. Shock-sensitive azide salts form in copper drain pipes, when exposed to sodium azide solutions.

Improperly sealed containers of toxic liquids, such as carbon tetrachloride and leaking cylinders of toxic gases, can poison the air.

Defective safety devices represent a category of insidious hazards.





For convenience, we will list insidious hazards in one of four categories. We will avoid specifically discussing dangers which are detailed elsewhere in the manual by referring to the appropriate sections. The categories, with some examples, are:

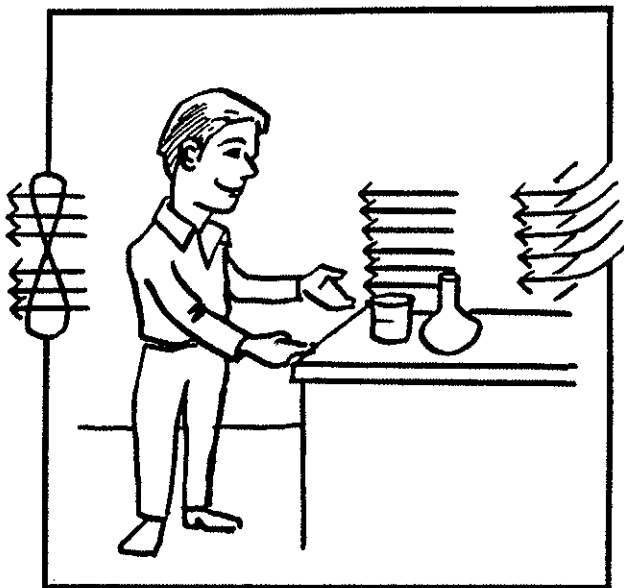
A. Unanticipated Chemical Reactions Due To:

1. Peroxide formation
2. Azide formation
3. Gas formation
4. Fulminate formation
5. Corrosion
6. An induction period
7. Many others

B. Unanticipated Toxicity (Chapter 3)

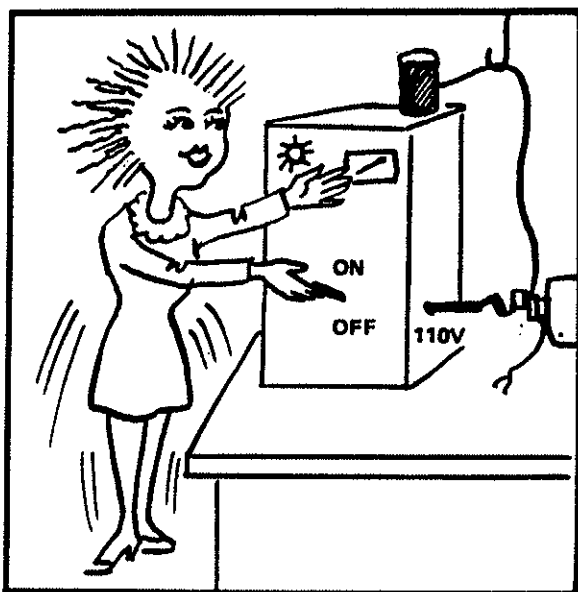
1. Carcinogen
2. Mutagen
3. Teratogen
4. Systemic poison
5. Others





C. Improper Facility Design

1. Hoods
2. Ventilation
3. Safety equipment
4. Egress
5. Alarms
6. Sprinkler systems
7. Safety showers/eye wash stations
8. Others



D. Improper Handling of Equipment and Chemicals

1. Protective safety equipment
2. Electrical grounding
3. Static electricity
4. Inspections/Inventory (Chapter 10)
5. Storage (Chapter 16)
6. Others

ALL DANGERS ARE INSIDIOUS IF YOU ARE UNAWARE OF THEM—CONTINUE YOUR EFFORTS TO UNCOVER ALL INSIDIOUS HAZARDS! ! !



ETHERS**DANGER!****CAUSES IRRITATION****HARMFUL IF SWALLOWED OR INHALED****EXTREMELY FLAMMABLE-
HIGHLY VOLATILE****TENDS TO FORM EXPLOSIVE PEROXIDES-
ESPECIALLY WHEN ANHYDROUS****AZIDES****Unusual Fire & Explosion Hazards****Can react violently with shock, friction or heat.****NOTE: Decomposes at melting point****Toxic Gases Produced****ammonia, nitrogen oxides,
hydrazoic acid fumes (explosion hazard)****A. Unanticipated Chemical Reactions**

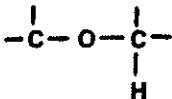
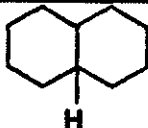
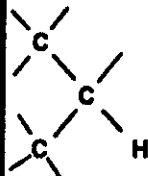
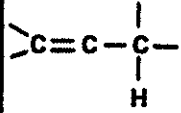
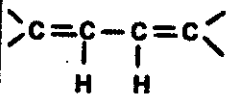
1. Peroxides may be formed in all of the compounds listed on the next two pages. The resultant peroxide may explode or initiate a violent polymerization reaction. Extreme care should be taken in handling peroxidizable compounds. Never store them longer than the manufacturer suggests. At a minimum, stock should be inventoried once a year and old supplies either discarded or professionally repurified. (The latter is dangerous, but can be done safely by an experienced chemist who takes proper precautions.) Do not depend on inhibitors to prevent peroxide formation indefinitely. They too have a limited lifetime. Test strips are available from the major chemical supply houses which allow peroxides to be determined quickly and efficiently.

2. Explosive lead and copper azides may form in drains, constant-temperature baths, reactor vessels, etc., if azide-containing solutions come in contact with these metals.

Consider alternate chemicals to replace azides.



POTENTIALLY DANGEROUS PEROXIDIZABLE COMPOUNDS

Structural Moiety ¹	Class	Compound	Synonyms
	Ethers	Diethyl Ether	Ethyl Ether Ethyl Oxide Ether Sulfuric Ether
		Diisopropyl Ether	Isopropyl Ether 2-Isopropoxy Propane
		Ethyl Vinyl Ether	Ethoxyethene EVE Vinamar
		Dibenzyl Ether	Benzyl Ether Benzyl Oxide
		Diallyl Ether	Allyl Ether
	Ether-like	2,2-Dimethoxypropane	
		Diethyl Acetal Acetaldehyde	Acetal 1,1-Diethoxyethane Ethylidene Diethyl Ether Diethylacetal
		Tetrahydrofuran	THF Diethylene Oxide Tetramethylene Oxide
	Decahydronaphthalenes	Decahydronaphthalene	DEC Decalin Napthanane
	Isopropyl Compounds	3-Isopropoxypropionitrile	
	Allyl Compounds		
	Dienes	Vinylidene Chloride	
		Dicyclopentadiene	3a,4,7,7a-Tetrahydro-4,7-methanoindene
		Butadiene	1,3-Butadiene, Divinyl
		2-Chloro-1,3-butadiene	Chloroprene

¹The H represents the peroxidizable hydrogen in the organic molecules.

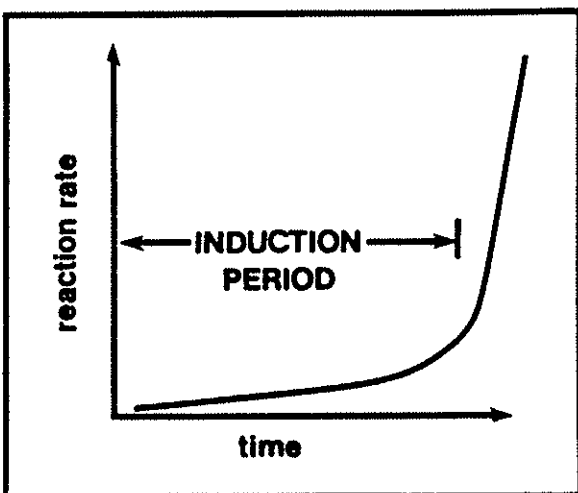
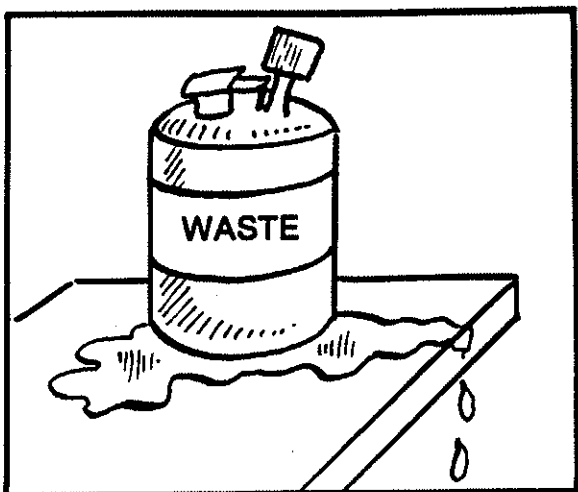
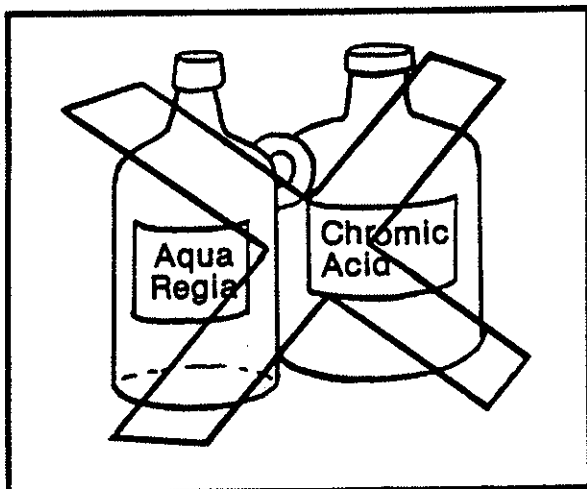


POTENTIALLY DANGEROUS PEROXIDIZABLE COMPOUNDS (Continued)

Structural Moiety ¹	Class	Compound	Synonyms
$\begin{array}{c} \diagup \\ \text{C}=\text{C} \\ \diagdown \quad \text{H} \end{array}$	Monomers	Acrylonitrile Acrylic Acid Butadiene Styrene Vinyl Acetate Vinyl Pyridine	
	Cyclic Enes	Cyclohexene 3,3,5-Trimethyl- 2-cyclohexen-1-one Cyclooctene	Isophorone
$\begin{array}{c} \diagup \\ \text{C}=\text{C} \quad \text{X} \\ \diagdown \quad \text{H} \end{array}$ or $\begin{array}{c} \text{X} \\ \diagup \\ \text{C}=\text{C} \\ \diagdown \quad \text{H} \end{array}$	Halogens Substituted on Alkanes	1,1-Dichloroethylene	
$\begin{array}{c} \diagup \\ \text{C}=\text{C}-\text{C}\equiv\text{C}- \\ \\ \text{H} \end{array}$	Vinylacetylenes	Vinylacetylene	
K, Rb, Cs	Alkali Earth Metals	Potassium Rubidium Cesium	
M(NH ₂)	Amides	Potassium Amide Sodium Amide	
...	Organometallics	Bis(-benzene)chromium(0) Bis(2,4-pentanedionato)- chromium	

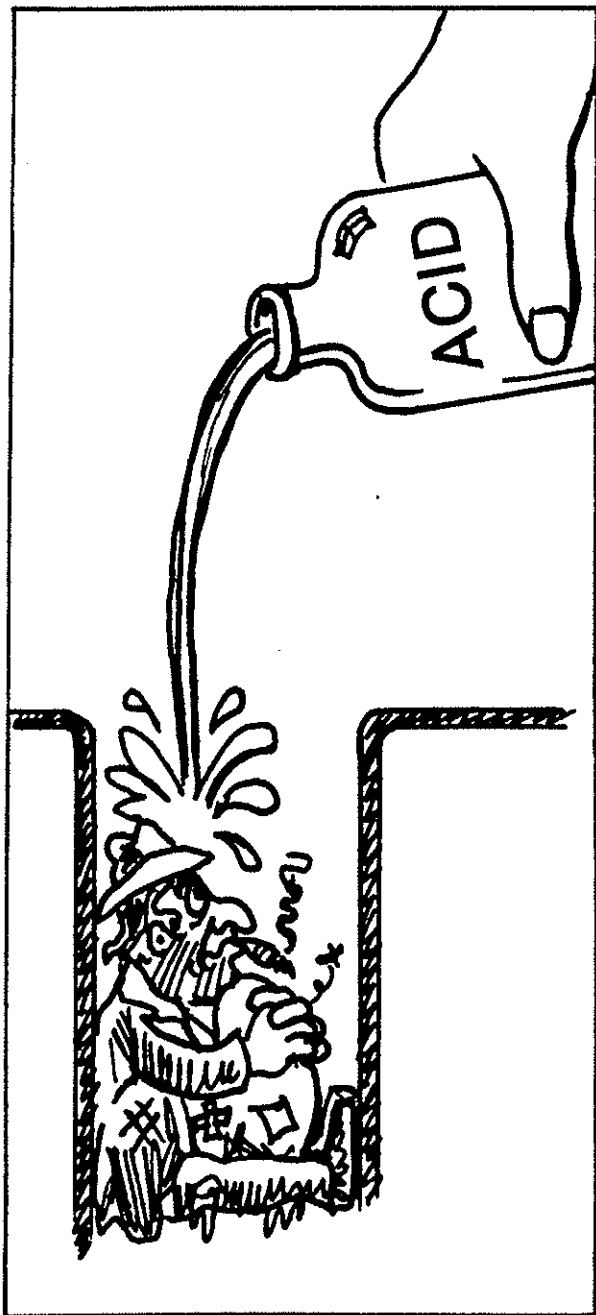
¹The H represents the peroxidizable hydrogen in the organic molecules.





3. Gas formation occurs as many oxidizing acids age. Nitric acid bottles should be vented periodically to avoid pressure build-up. Aqua regia and chromic acid should never be stored in tightly sealed containers. Many other compounds gradually decompose liberating gas (e.g., 2-chloromethylthiophene)
4. Silver, lead, and mercury in the presence of ammoniacal solutions can form explosive fulminates. Avoid silver and lead solders in reactors which will come into contact with aqueous ammonia. Dispose of Tollen's reagent immediately after preparation and use. Do not store it! Clean glassware which comes in contact with Tollen's reagent before it dries to form the explosive silver fulminate.
5. Corrosion of glass may occur with the inadvertent production of HF during a reaction. Mixtures of chlorinated solvents with THF and water can corrode stainless steel, thus destroying waste solvent cans.
6. Reactions with induction periods appear to start slowly then take off. Research all reactions thoroughly and avoid excessively large scale-ups.





7. In the laboratory, one common source of insidious hazards is the sink drain! If aqueous solutions, such as test and reaction mixtures, are disposed of by flushing down the drain, this can lead to the build-up of toxic or otherwise hazardous materials, awaiting a catalyst, perhaps an acid, to release them into the laboratory air. When flushing down the drain, use copious quantities of water; don't flush reactive chemicals at the same time. Don't flush down chemicals capable of releasing toxicants to the environment.





B. Unanticipated Toxicity

Most of these problems are discussed in Chapter 3 of this manual.

One of the most common insidious hazards is mercury. Mercury is widely used in such diverse items as electric switches, amalgams, boilers, barometers, thermometers, lamps, and cells. In the laboratory, mercury is commonly found in thermometers, barometers, and certain analytical instruments. Additionally, mercury compounds are common reagents found on nearly all laboratory stock shelves.

Because of its widespread use, the hazardous nature of mercury may be overlooked or ignored, even when the hazard is understood. Common treatment for a mercury spill, say a broken thermometer, is to aspirate any visible drops. However, many small droplets may be hidden in small cracks and crevices and are left to volatilize into the atmosphere.

The Mercury Hazard

Physical Properties: Mercury is a very dense liquid. It is capable of forming stable, very finely divided droplets. The vapor pressure at 0°C: 0.000185 mm Hg.

Flammability: Non-flammable, but can form explosive compounds, such as mercury fulminates.

TWA: 0.05 mg/m³

The saturated concentration of mercury vapor at room temperature is about 20 mg/m³ or 400 times the TWA.

One milliliter, less than the amount in an average laboratory thermometer, can increase the mercury level of millions of cubic feet of air to above the TWA.

Effects:

Acute: Burning of throat and mouth, shock, cardiac arrest, nausea, vomiting, abdominal pain, bloody diarrhea, difficult breathing, bronchitis, inflammation of the lungs.

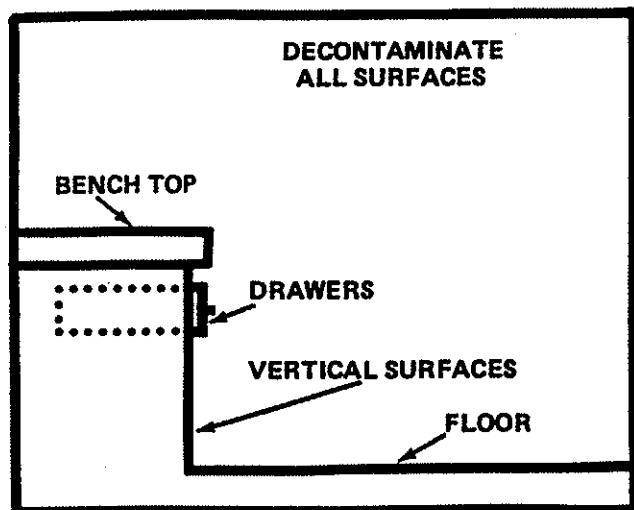
Chronic: CNS: headache, dizziness, restlessness, inflammation of nerves, loss of muscle control, tremors.

Gastrointestinal: Inflammation of gums, nausea, vomiting, diarrhea, liver damage.

Respiratory: Inflammation of nose and lungs, cough, fever.

Erethism: Irritability, outbursts of temper, excitability, shyness, resentment of criticism, headache, fatigue and indecision.





Mercury droplets (10-100 micrometer diameter in size) adhere to vertical surfaces and penetrate into porous flooring. Large amounts of mercury can be left undiscovered after spills.

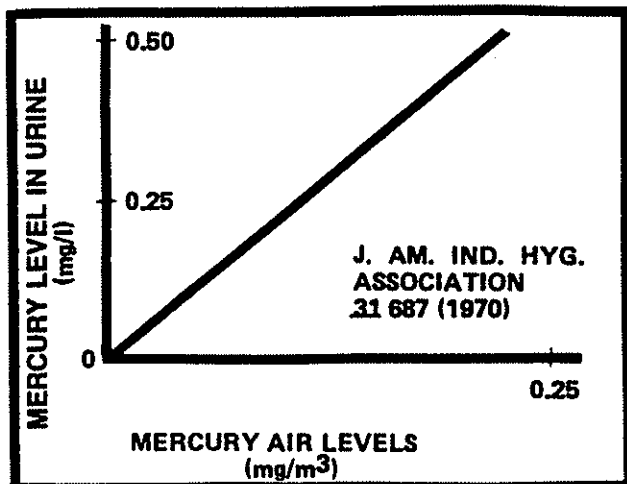
Unless spills are promptly cleaned up and the area decontaminated, the build-up will continue. All exposed surfaces must be thoroughly decontaminated.

MERCURY VAPOR

t°F	t°C	V.P.*	mg/m ³	$\frac{\text{mg/m}^3}{\text{TWA}}$
50	10	0.4	5.9	120
68	20	1.2	13.2	260
75	24	1.7	18.3	370
82	28	2.4	25.2	500
97	36	4.5	46.6	930
104	40	6.1	62.6	1300
122	50	12.7	131	2600
212	100	273	3000	60000

*X10⁻³ mm Hg

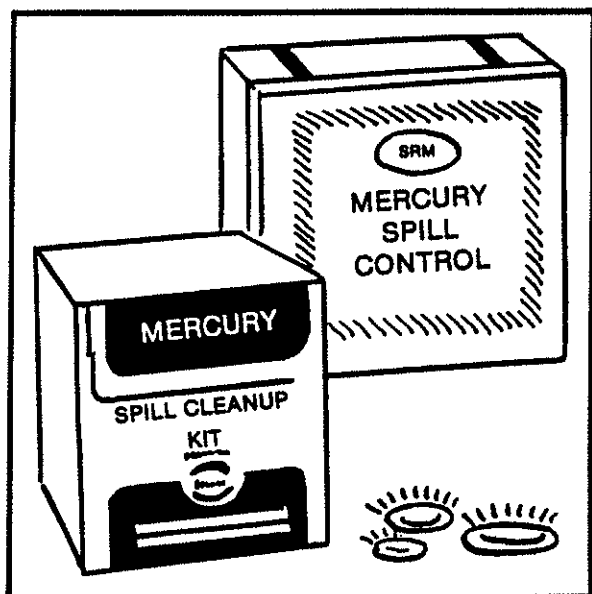
Mercury spilled on hot surfaces or in ovens will quickly vaporize.



At concentrations up to 5 mg/m³ a type 1 respirator (full facepiece gas mask equipped with a high efficiency filter plus a canister containing iodine-impregnated charcoal) can be used.

Refer to the NIOSH publication *Occupational Exposure to Inorganic Mercury and The Environmental Mercury Problem* by F. M. D'Itri, CRC Press, 1972.





All mercury spills should be cleaned up quickly and efficiently. To minimize the difficulty of cleaning up a spill and decrease the associated hazards, mercury-containing devices should always be in appropriate catch pans. Spill cleanup kits are available. Monitoring devices should be used. (See Chapter 4)

General cleanup procedures follow.

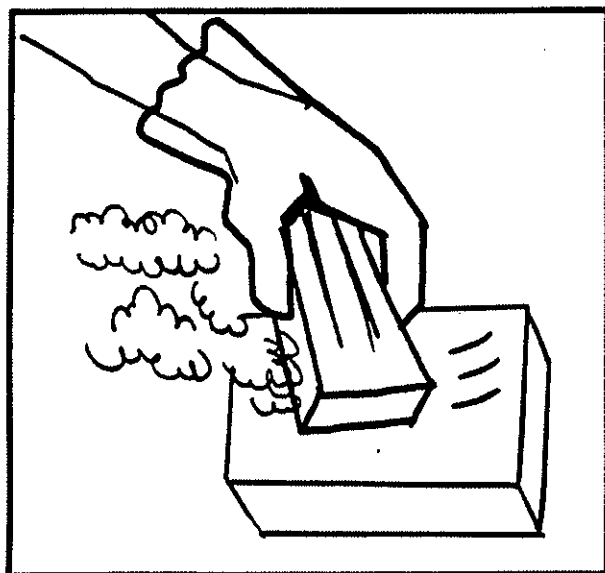
First, always turn off any hot equipment in the spill area.

Second, don proper protective wear:

gloves a must

laboratory coat or apron recommended

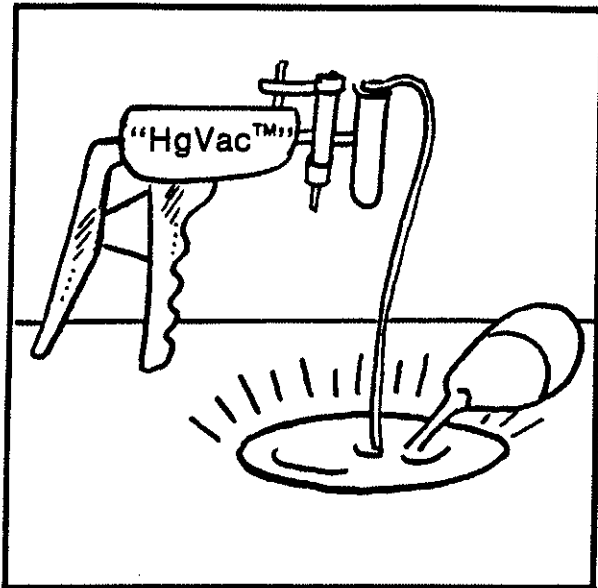
respirator a must if
mercury is on
a hot surface
(simple face
masks do little
to protect you
from mercury
vapors)



Third, increase ventilation in the area.

Fourth, clear the area of clutter. Check and clean all surfaces as you remove these items. Be sure to remove any gold, silver, or copper which may amalgamate with the mercury. Small droplets on vertical surfaces of the items being removed may be cleaned up with a mercury sponge.

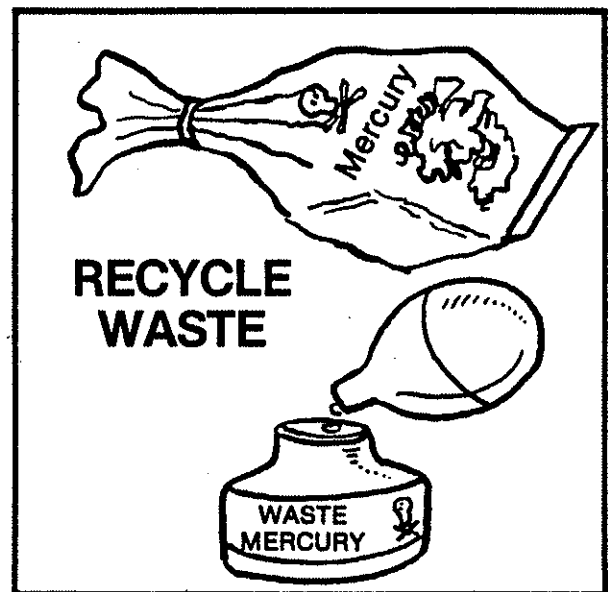
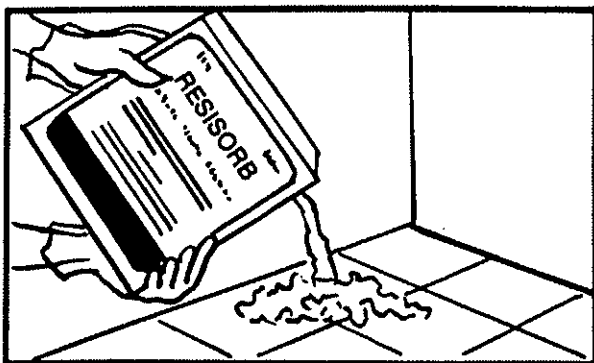
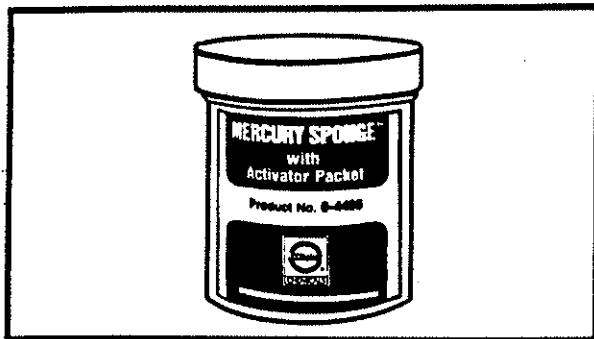




Fifth, push mercury together and mechanically retrieve it. Put it into a waste mercury bottle.

Sixth, absorb residual mercury with a sponge designed for this purpose, or use a mercury vapor absorbent to absorb vapors from mercury which may be in inaccessible areas such as cracks and crevices.

Finally, return all waste mercury and mercury sponges to the manufacturer for recycling. Do not dispose of mercury or mercury-containing material in any way. Recycle it!



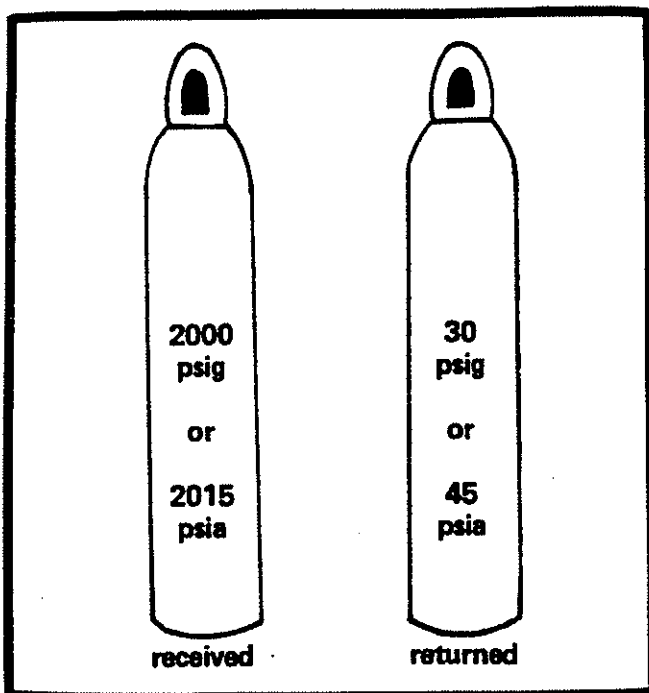


A summary of control measures for insidious hazards is given below:

1. Prepare a checklist of insidious hazards. Use the NFPA rating as a guide. A value of 3 or 4 indicates a serious hazard.
2. Thoroughly train all operators in the hazards of substances commonly used in the laboratory.
3. Provide adequate ventilation in the form of hoods and forced air.
4. Do not allow stock build-up of toxic, flammable, or corrosive materials.
5. Have efficient cleanup agents available for spills.
6. Have suitable disaster equipment available (fire extinguishers, gas masks).
7. Secure periodic medical examinations to detect insidious chemicals in body tissues.
8. Include an insidious hazard section in your safety audit.

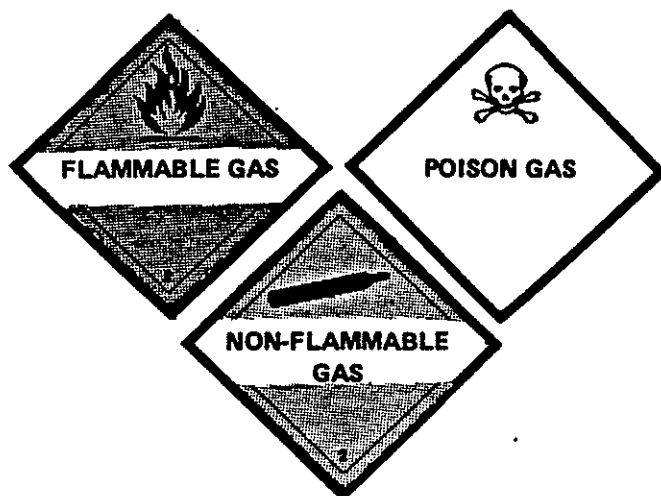


13. Compressed Gases



$$\text{psia} = \text{psig} + 15 \text{ psi}$$

$$\left\{ \begin{array}{l} \text{pounds per} \\ \text{square inch} \\ \text{absolute} \end{array} \right\} = \left\{ \begin{array}{l} \text{pounds per} \\ \text{square inch} \\ \text{gauge} \end{array} \right\} + \left\{ \begin{array}{l} 15 \text{ pounds} \\ \text{per square} \\ \text{inch} \end{array} \right\}$$



Compressed gases are defined by DOT as those nonflammable materials which have a psia greater than 40 at 70°F or a psia greater than 104 at 130°F, or those flammable materials which have a psia greater than 40 at 100°F.

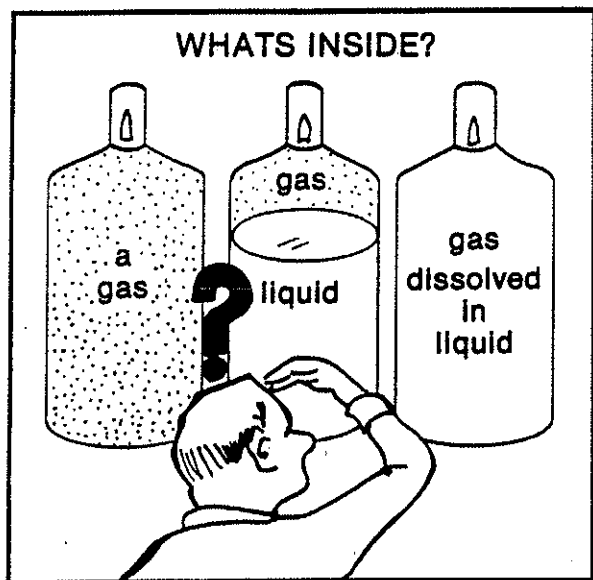
A compressed gas is considered flammable if at least one of the following is true.

1. The lower flammability limit is less than 13% v/v.
2. The flammability range is greater than 12%.
3. Flame is 18 inches or longer in the standard Bureau of Explosions test apparatus.
4. The open drum test apparatus propagates the flame.
5. The closed drum apparatus explodes.

A compressed gas is considered nonflammable if none of the above applies.

Those with sufficiently high toxicity are classified as Class A poisons and receive a poison gas label.

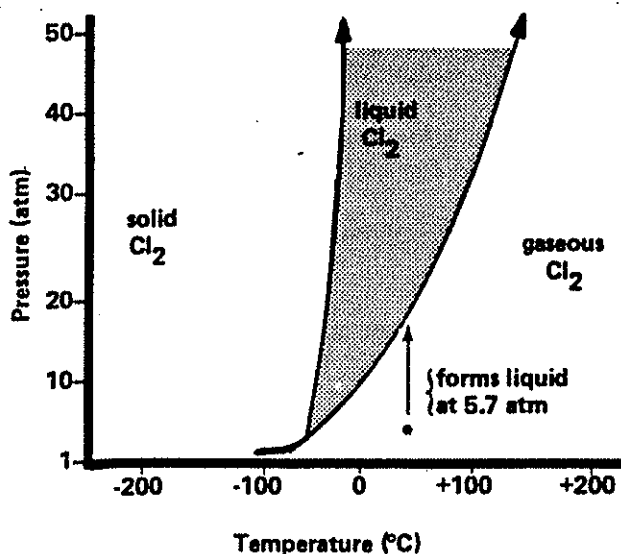




We need to be especially aware of:

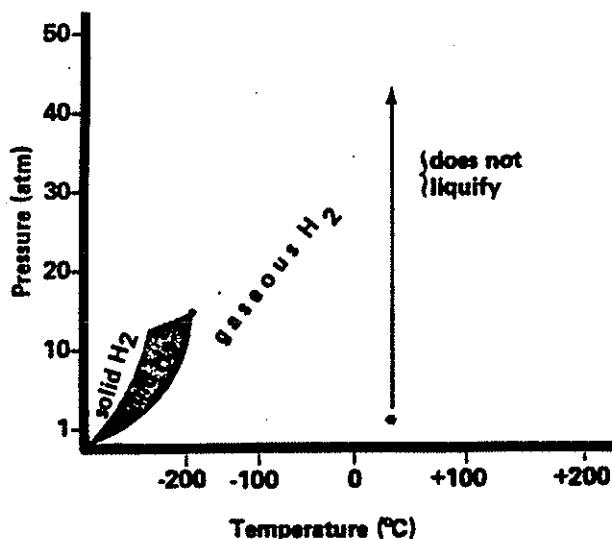
1. The physical state of the material inside the cylinder.
2. The chemical hazards of the gas.
3. The physical hazards of the cylinder itself.
4. The proper safety procedures involved in using the materials.

Compressed gases are stored in heavy-walled metal cylinders which are specially constructed, maintained, and tested for such use. Actual cylinder contents may be liquified gas, non-liquified gas (i.e. compressed gas), a gas dissolved in a liquid or solid, or a gas mixture.

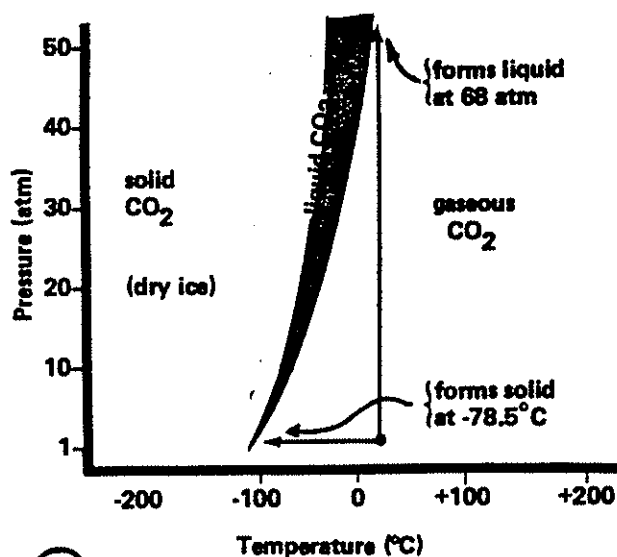


Chlorine, Cl_2 , exists in the cylinder as a liquid, but at normal pressure, 1 atmosphere or 15 psia, and temperature, 25°C , it is a gas. The phase diagram to the left shows how chlorine can be liquified by increasing the pressure.





Hydrogen, H_2 , cannot be liquified at room temperature by applying pressure. It exists in the cylinder as a gas under pressure.



Carbon Dioxide, CO_2 , can be liquified at room temperature by increasing pressure to 68 atmospheres. It can be solidified by cooling it to $-78.5^{\circ}C$. This is dry ice.



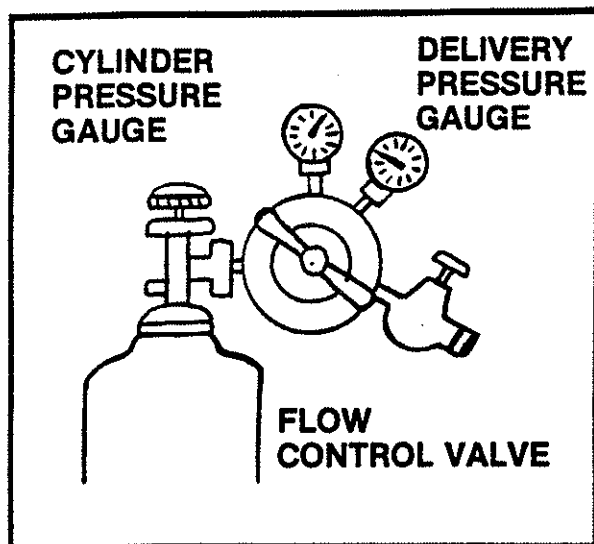
300 parts acetylene gas dissolved in
1 part acetone liquid at 175 psia

Acetylene is supplied as a gas dissolved in a solvent, frequently acetone. Other gases, such as diborane, are diluted with inert gas for safety.

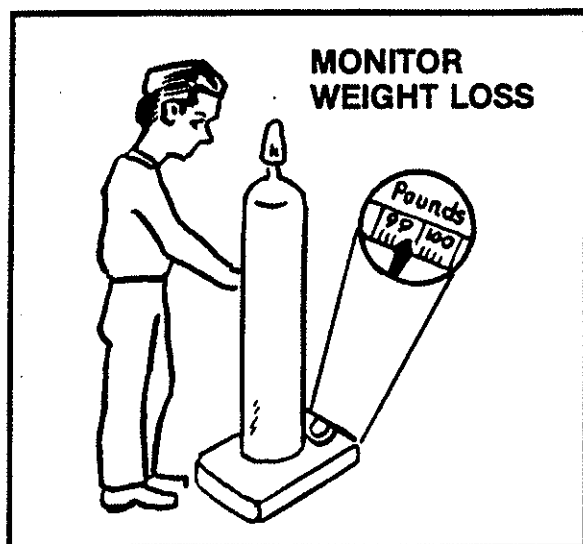
The physical state of a variety of materials contained in typical compressed gas cylinders is noted in the first four columns of the table on page 13-7.



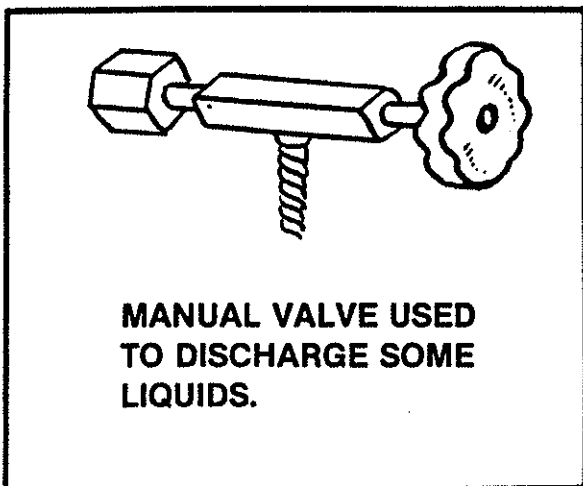
© J.T. Baker Inc.



The amount of material remaining in a non-liquified gas cylinder or a dissolved gas cylinder is directly proportional to the pressure inside the cylinder, the cylinder pressure gauge reading. Cylinders should be returned to the manufacturer before the gauge pressure falls to zero.

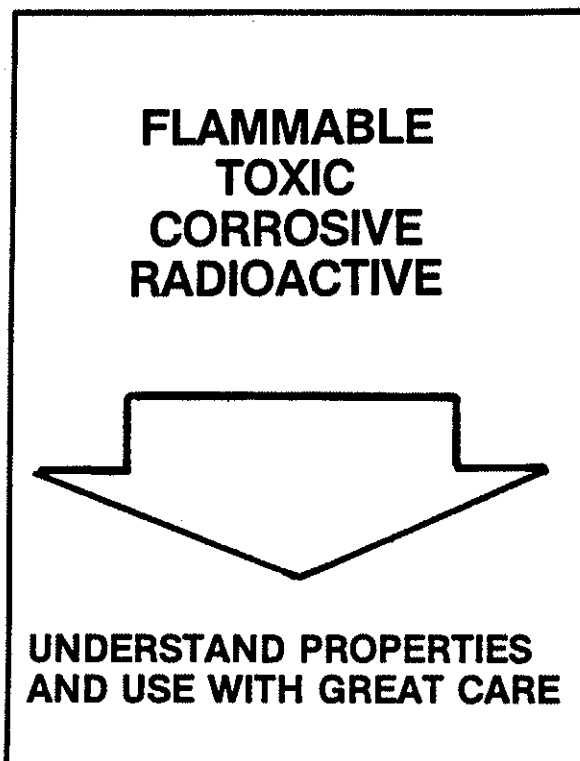


The amount of liquified gas remaining in a cylinder can be determined only by weighing it. The net weight of the contents should be noted on a card attached to the cylinder. The net weight of the cylinder itself is stamped on the neck or valve stem.



Regulators are meant to control delivery gas pressure. Do not use a regulator on a liquified gas cylinder when the cylinder is on its side because liquid will contact the regulator. Special discharge devices are needed to remove liquid from the cylinder.

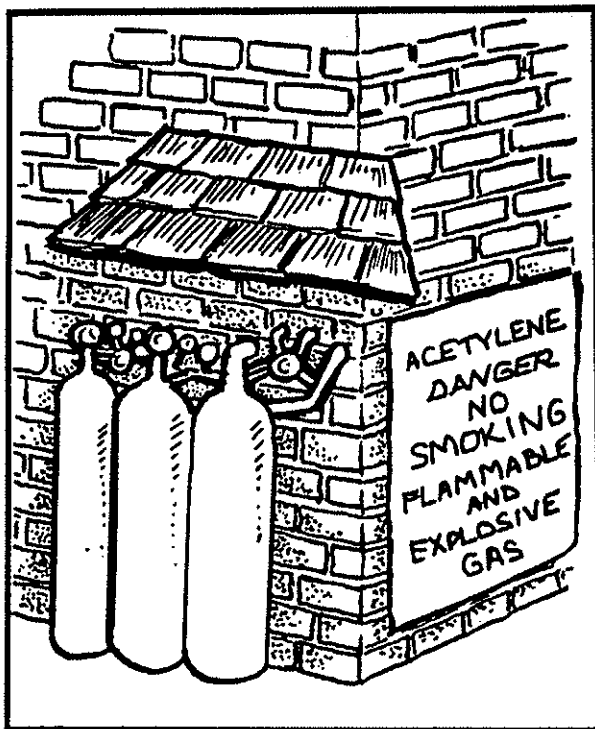




Chemical Hazards

Chemical dangers exist for all the gases. System leaks can very quickly disperse dangerous chemicals throughout the environment.

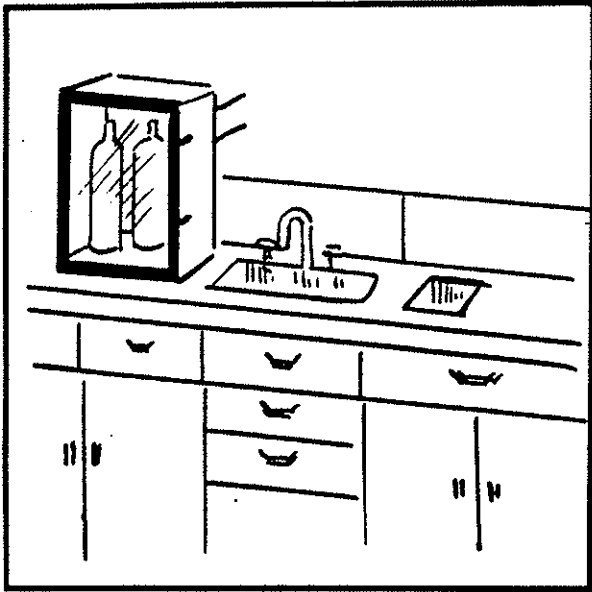
All flammable compressed gases, liquified, non-liquified, and dissolved, have flash points below room temperature. It is impossible to avoid forming a flammable mixture if they leak into the air.



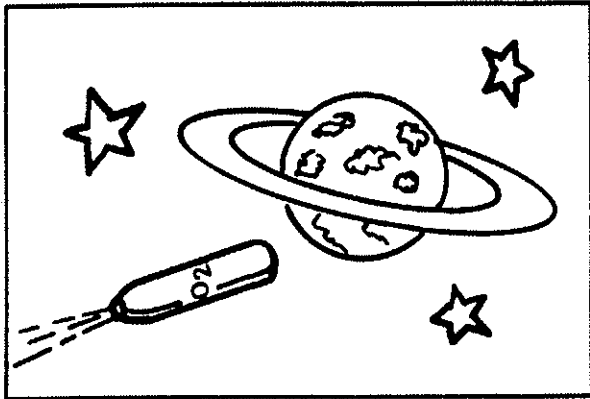
Read and understand the dangers of the material you are working with!

Use flammables in a well-ventilated area. Never exceed the NFPA guideline limiting the storage of flammables to two cylinders per laboratory. Preferably, store flammables outside and pipe them in.





Limit the quantity of toxic gases available. Use them in a well-ventilated area and, in the case of extremely toxic materials such as phosgene, seal the area and handle the process by remote control. Enter only with an adequate self-contained breathing apparatus. Our best predictions indicate that the release of a mere 200 pounds of phosgene could kill everyone within one-quarter mile of the site.



Corrosive gases can attack gauges, plumbing, and hood areas. Make sure valves on corrosive gas cylinders are fully opened when the gas is used and that they are turned open and shut at least once a day. This avoids freezing of the valve.

Oxygen under pressure and in contact with organic matter can cause a severe explosion or fire. Never allow organic matter to collect in regulators or lines carrying oxygen or compressed air.



CHARACTERISTICS OF SOME COMMON COMPRESSED GASES

CHEMICAL	PHYSICAL STATE WITHIN CYLINDER				BOILING AND FREEZING POINTS	CRITICAL TEMPERATURE AND PRESSURE		D.O.T. CLASS	SAFETY RELIEF DEVICE				TOXICOLOGICAL AND FLAMMABILITY DATA					
Cylinder Contents	Pressurized Gas	Pressurized Liquid	Dissolved Gas	Diluted with Gas	Boiling Point (°C)	Freezing Point (°C)	Critical Point (atmospheres)	Critical Point (°C)	Class A Poison	Flammable Gas	Nonflammable Gas	Corrosive Liquid	Fusible Disk	Frangible Disk	Spring-loaded	Combination, etc.	TLV® or PEL	Flammability Limits (%) In Air
Acetylene			x		-71		62	36		x			x				?	2.5-81
Allene		x			-36	-137	54	120		x							?	7-7
Ammonia	x	x			-33		112	132			x						50	16-28
Argon					-185	-190	48	-122						x				
Arsine				x	-62			100	x	x						x N	.05	7-7
Boron Trichloride		x			12.5	-107	38	179				x					?	
Boron Trifluoride	x	x			-100	-127	49	-12			x						1	
Bromine Pentafluoride		x			41	-80		197				x					0.1	
Bromotrifluoroethylene		x			-3					x			x					7-7
Bromotrifluoromethane		x			-58	-108	39	67			x			x				
1,3-Butadiene		x			-4	-109	43	152		x			x	x	x	x	1000	2-11.5
Butane		x			0	-138	37	152		x			x	x	x	x		1.9-8.5
Carbon Dioxide		x			subl	-78	73	31			x			x				
Carbon Monoxide	x				-191		34	-140		x				x			50	13-74
Carbon Tetrafluoride	x				-128	-184	37	-46			x			x				
Carbonyl Sulfide		x			-60	-139	61	105		x			x				10	12-29
Chlorine		x			-34	-150	76	144			x		x				1	
Chlorodifluoromethane		x			-41	-160	49	96			x		x				1000	
Chloropentafluoroethane		x			-39	-106	31	80			x		x				1000	
Chlorotrifluoroethylene		x			-28	-158	40	106		x			x				20	8-39
Chlorotrifluoromethane		x			-81	-294	38	29			x			x			1000	
Cyanogen		x			-21	-28	58	126	x	x						N	10	6-32
Cyclopropane		x			-33	-128	54	124		x				x		N	400	2-10
Diborane				x	-93	-185	40	17		x	x					N	0.1	1-98
Dimethylamine		x			7	-62	52	165		x							10	3-14
Dimethylether		x			-24	-142	53	129			x			x	x			3-18
Ethane		x			-88		48	32			x			x	x			3-13
Ethylene		x			-165	-273	52	10			x			x			5000	3-32
Fluorine	x				-188		55	-129			x					N	0.1	
Germane					-90	-185			x	x						N	0.5	7-7
Helium	x				-269	-272	2.3	-268			x			x				
Hydrogen	x				-253	-259	.13	-240		x						x		4-75
Hydrogen Chloride		x			-85	-114	81	51			x					x	5	
Hydrogen Cyanide		x			26	-13	53	184	x							N	10	6-41
Hydrogen Selenide		x			-42	-68	88	138	x	x						N	0.05	7-7
Hydrogen Sulfide		x			-77	-122	89	100		x			x				10	4-45
Methane	x	x			-161	-182	46	-82		x			x			x		5-14
Methyl Acetylene		x			-23	-102	55	129		x			x		x		1000	2-12
Methyl Chloride		x			-24	-96	66	143		x			x				100	8-17
Methyl Vinyl Ether		x			5.5	-122				x					x		?	3-39
Nickel Carbonyl		x		x	43	25	30	200		x						x	.001	7-7
Nitric Oxide	x				-151	-164	64	-93								N	25	
Nitrogen	x				-196		34	-147			x			x				
Nitrous Oxide		x			-90	-91	72	36			x			x				
Oxygen	x				-183		50	-118			x			x				
Ozone			x		-112	-192	55	-12			x						0.1	
Phosgene		x			8.2	-128	56	-182	x							N	0.1	
Phosphine		x			-88	-134	64	51								N	0.3	
Phosphorous Pentafluoride	x				-84	-94	125	15			x					x	3	
Propane		x			-42	-188	42	99		x			x	x	x		1000	2.2-9.5
Silane	x	x			-112	-185	48	-4		x						N	?	7-7
Sulfur Dioxide		x			-10	-75	78	168			x		x				5	
Vinyl Chloride		x			-14	-169	53	158		x				x	x		500	4-22

Notes: 'x' in any position means that the chemical belongs to that group.

'N' in any safety relief device column means that no relief device is allowed by D.O.T.

'' in any position a blank means that the chemical does not have those characteristics or requirements. All blanks in the safety relief device columns means that none is used. A blank in the toxicological column means a TLV or PEL has not been established, but the material is believed to be either non-toxic or mildly toxic.

A blank in the flammability limit column indicates that the gas is non-flammable.

'?' A question mark means that the danger is not known but it is or can be severe.

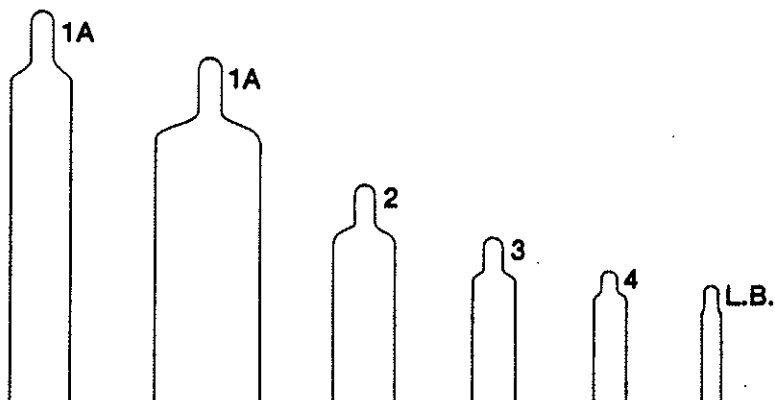


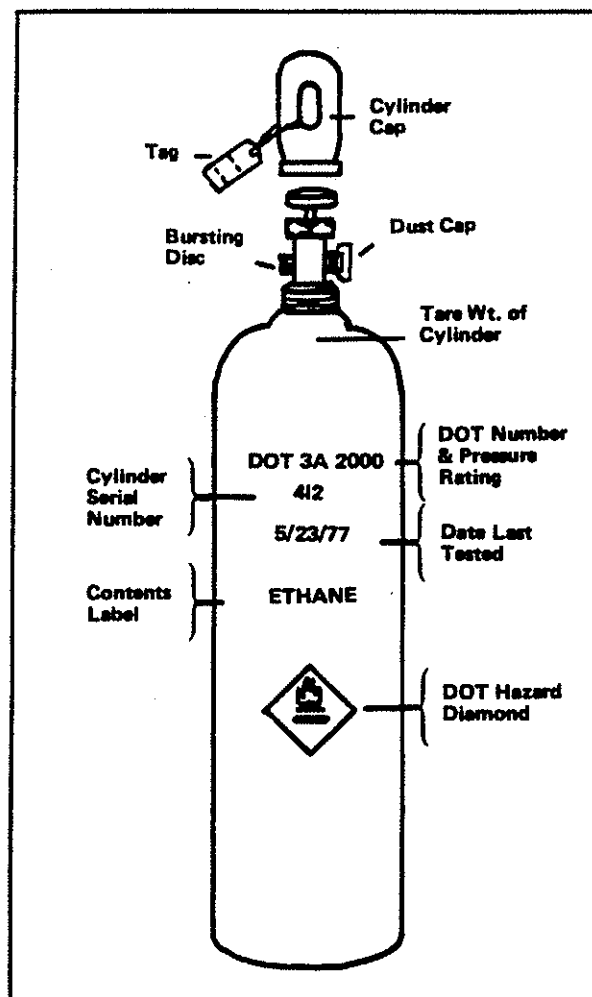
Guidelines for limiting the amount of compressed gas in any laboratory can be adapted from CGA Pamphlet P-1-1965. An example follows:

LABORATORY GAS CYLINDER LIMITATIONS

Description	Extremely Toxic	Very High Toxicity	Highly Toxic	Toxic	Liquified Flammable	Flammable	Inert
Examples	Arsine Fluorine Phosphine Silane	Boron Trichloride Chlorine Nickel Carbonyl	Carbon Monoxide Cyanogen Hydrogen Cyanide	Ammonia Hydrogen Chloride Sulfur Dioxide	Butane Propane Cyclopropane Propylene	Hydrogen Ethane Methane Ethylene	Argon Air (?) Carbon Dioxide Helium
Special Precautions	Must be stored in hood or vented closet and used in adequate hood.				—	—	—
Maximum Cylinder Size	Special	L.B.	No. 4	No. 4	No. 2	No. 1A	No. 1A
Maximum Number of Cylinders in One Class in One Lab	Special Consideration			2 2 L.B.	2 15 L.B.	2 15 L.B.	10
Total Number of Cylinders in One Lab	4						10

RELATIVE COMPRESSED GAS CYLINDER SIZES





The cylinder itself should be received in good condition with properly identified and fitted components as indicated to the left. Cylinder caps must always be in place except when the cylinder is in service.

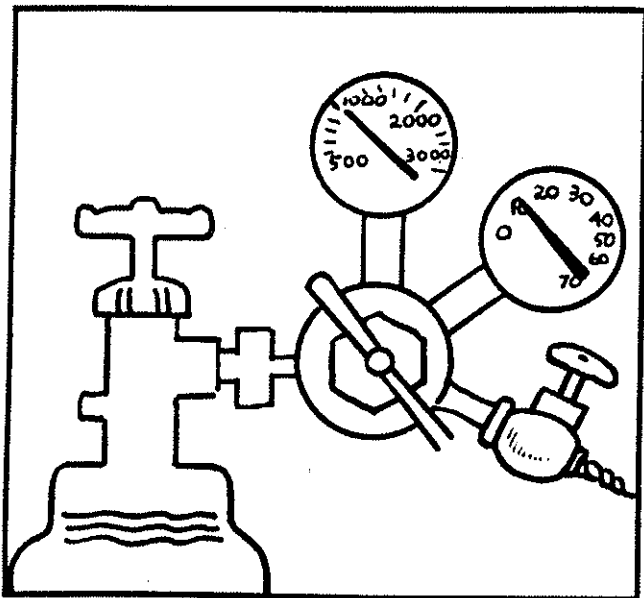
A tag should identify the contents and warn of any special dangers. Color coding is not a standardized means of identification and should never be used as such. Content markings on the shoulder of the cylinder comply with OSHA 1910.252 and ANSI Z 48.1-1954.

Except for Class A poisons, fluorine, and lecture bottles, safety devices specified by DOT protect the cylinder from rupturing due to excess heat and/or pressure. Safety devices are located on the bottom of acetylene cylinders. Never tamper with any safety device.

A DOT hazard diamond is required for the interstate transportation of cylinders.

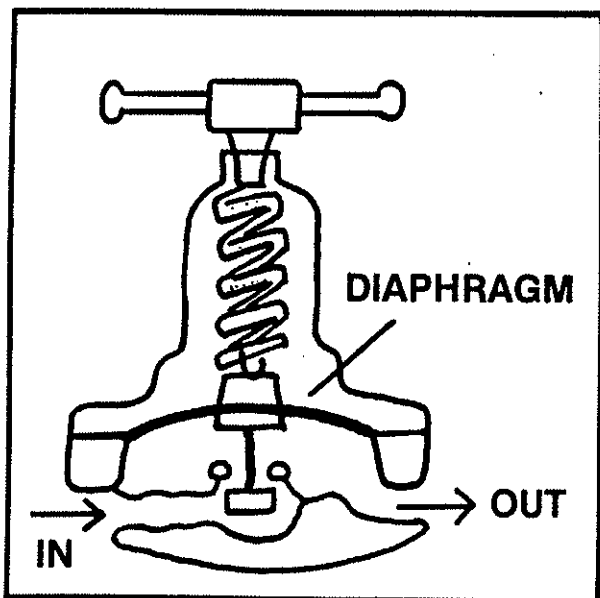
Hydrostatic testing is prescribed by DOT regulations every five or ten years. The longer interval is indicated by a five-pointed star stamped after the test date on the cylinder. See 49 CFR 173.34 for specific details.





Transport the cylinder to the service area in a secure cart with the cylinder cap in place. Secure the cylinder. Remove the cylinder cap and the dust cap. Connect the proper regulator. Do not use adaptors to effect a connection because dangerous reactions may occur with contaminants or the regulator itself.

Cylinders which are not in daily service should have their regulator removed and dusted; the cylinder cap should be replaced after each use.



Regulators are diaphragm devices designed for pressure control only. Do not attempt to control flow with the diaphragm regulator. Pressure is increased by turning the hand valve stem clockwise.



Rules for Handling Compressed Gases

1. Always use a hand truck for transport. Chain cylinder to hand truck.
2. Do not transport in closed vehicles.
3. Cylinders should be chained in place or otherwise secured at all times.
4. Do not drop cylinders, or otherwise permit them to strike each other.
5. Leave valve cap on cylinder until secured and ready for use.
6. Ground all cylinders containing flammable gases.
7. Use only in an upright position.
8. All valves should be closed when not in actual use.
9. Use the proper regulator for the particular gas.
10. Carefully open all valves and adjust gas flow rates.
11. Always consider cylinders to be full and handle accordingly.
12. Discontinue using a high-pressure cylinder when the pressure approaches 30 psig, and clearly mark EMPTY; then remove for return to vendor.
13. Oily (not specially cleaned) fittings should never be used with oxygen. Oxygen under pressure will rapidly oxidize oil or grease, resulting in an explosion. Equipment cleaned for oxygen service must be used.
14. Acetylene under pressure can decompose with explosive force. It can explode with extreme violence if ignited. Copper or brass (with more than 65% copper) can form explosive compounds in contact with acetylene.
15. Glass equipment should not be pressurized. A general rule is no pressure greater than 10 inches of water, without special protective equipment.
16. Never mix gases in a cylinder. Explosion, contamination, corrosion, and other hazards can result.
17. Cylinders containing large amounts of a flammable gas (hydrogen, acetylene, ethylene) should be stored outside in a protected area and piped into the working area.
18. Store in a fire-proof, well-ventilated area.
19. Storage area temperature should be regulated, so as not to exceed 100° F.
20. Store gases supporting combustion (O_2 , Cl_2 , etc.) at least 25 feet from fuel gases, preferably in another gas storage area.

